

Root data
SGA 3, Exposé XXI

English draft

M. Demazure

About this draft

This is an unofficial English translation of Exposé XXI, “Root data” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp21-13oct24.pdf` (46 pages), from the source collection. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XXI. Root data

by M. Demazure

⁽⁰⁾This exposé collects, in the absence of a suitable reference,⁽¹⁾ known results on root data (= “abstract” root systems), most of which will be used in what follows.

Notations. One denotes by $\mathbf{Q}_+$ the set of positive (or zero) rational numbers; one has $\mathbf{Z} \cap \mathbf{Q}_+ = \mathbf{N}$. Let V be a $\mathbf{Q}$ -vector space; if A (resp. B) is a subset of $\mathbf{Q}$ (resp. V), one denotes by $A \cdot B$ the image of $A \otimes B$ by the morphism $\mathbf{Q} \otimes V \rightarrow V$, in other words the set of linear combinations of elements of B with coefficients in A . One puts $-B = \{-1\}B$. One denotes by $E - F$ the set of elements of E which do not belong to F .

1. Generalities

1.1. Definitions, first properties

Definition 1.1.1. Let M and M^* be two free $\mathbf{Z}$ -modules of finite type in duality. Put $V = M \otimes \mathbf{Q}$, $V^* = M^* \otimes \mathbf{Q}$; these are two $\mathbf{Q}$ -vector spaces in duality. Identify M (resp. M^*) with a subset of V (resp. V^*). The canonical bilinear form on $M^* \times M$ (resp. $V^* \times V$) is denoted $(\ , \)$.

Let R be a finite subset of M . Give a map $\alpha \mapsto \alpha^*$ from R to M^* ; the set of α^* for $\alpha \in R$ is denoted R^* . With each $\alpha \in R$, associate the endomorphism s_α (resp. s_α^*) of M and V (resp. M^* and V^*) given by the formulas:

$$(1) \quad s_\alpha(x) = x - (\alpha^*, x)\alpha, \quad \text{i.e. } s_\alpha = \text{id} - \alpha^* \otimes \alpha;$$

$$(1^*) \quad s_\alpha^*(u) = u - (u, \alpha)\alpha^*, \quad \text{i.e. } s_\alpha^* = \text{id} - \alpha \otimes \alpha^*.$$

One says that the pair (R, R^*) (more precisely the pair $(R, R \rightarrow M^*)$) is a *root datum* in (M, M^*) , or that (M, M^*, R, R^*) is a root datum, if the following axioms hold:

(DR I) For each $\alpha \in R$, one has $(\alpha^*, \alpha) = 2$.

(DR II) For each $\alpha \in R$, one has $s_\alpha(R) \subset R$, $s_\alpha^*(R^*) \subset R^*$.

One says that R is the *root system* of the root datum $\mathcal{R} = (M, M^*, R, R^*)$. The elements of R (resp. R^*) are called the *roots* (resp. *coroots*) of the root datum.

Remark 1.1.2. Axiom (DR I) is equivalent to any one of the following properties:

$$\begin{aligned} (2) \quad s_\alpha s_\alpha &= \text{id}, & (2^*) \quad s_\alpha^* s_\alpha^* &= \text{id}, \\ (3) \quad s_\alpha(\alpha) &= -\alpha, & (3^*) \quad s_\alpha^*(\alpha^*) &= -\alpha^*. \end{aligned}$$

Remark 1.1.3. Axioms (DR I) and (DR II) imply

$$R = -R, \quad R^* = -R^*, \quad 0 \notin R, \quad 0 \notin R^*.$$

Lemma 1.1.4. The map $R \rightarrow R^*$ is a bijection. More generally, if $\alpha, \beta \in R$ and $(\alpha^*, x) = (\beta^*, x)$ for every $x \in R$, then $\alpha = \beta$.

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** For the results on root systems (§§ 1–5), one may consult [BLie], Chap. VI.

Indeed, one then has $s_\beta(\alpha) = \alpha - 2\beta$, $s_\alpha(\beta) = \beta - 2\alpha$. One deduces at once

$$\begin{aligned} s_\beta s_\alpha(\alpha) &= 2\beta - \alpha = \alpha + 2(\beta - \alpha), \\ s_\beta s_\alpha(\beta - \alpha) &= s_\beta(\beta - \alpha) = \beta - \alpha, \end{aligned}$$

whence $(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha) \in R$ by (DR II). Since R is finite, one has $\beta - \alpha = 0$.

Corollary 1.1.5. *The inverse map $R^* \longrightarrow R$ defines a root datum*

$$\mathcal{R}^* = (M^*, M, R^*, R)$$

called the dual of $\mathcal{R}$.⁽²⁾

Definition 1.1.6. Denote by $\Gamma_0(R)$ the subgroup of M generated by R . Denote by $\mathcal{V}(R)$ the vector subspace of V generated by R , that is, $\Gamma_0(R) \otimes \mathbf{Q}$. Applying these definitions to $\mathcal{R}^*$, one similarly constructs $\Gamma_0(R^*)$ and $\mathcal{V}(R^*)$.

One calls the *reductive rank* of $\mathcal{R}$ the number

$$\text{rgred}(\mathcal{R}) = \text{rank}(M) = \dim(V) = \dim(V^*) = \text{rank}(M^*) = \text{rgred}(\mathcal{R}^*).$$

One calls the *semisimple rank* of $\mathcal{R}$ the number

$$\text{rgss}(\mathcal{R}) = \text{rank}(R) = \text{rank}(\Gamma_0(R)) = \dim(\mathcal{V}(R)).$$

One therefore has $\text{rgss}(\mathcal{R}) \leq \text{rgred}(\mathcal{R})$.

We shall see below that $\text{rgss}(\mathcal{R}) = \text{rgss}(\mathcal{R}^*)$, i.e. that $\mathcal{V}(R)$ and $\mathcal{V}(R^*)$ have the same dimension.

Definition 1.1.7. One says that $\mathcal{R}$ is *semisimple* (resp. *trivial*) if $\text{rgss}(\mathcal{R}) = \text{rgred}(\mathcal{R})$ (resp. $\text{rgss}(\mathcal{R}) = 0$). For $\mathcal{R}$ to be trivial, it is therefore necessary and sufficient that R be empty. The trivial root datum of reductive rank zero is denoted $0 = (\{0\}, \{0\}, \emptyset, \emptyset)$.

Definition 1.1.8. Denote by $W(\mathcal{R})$ the group of transformations of M generated by the s_α , $\alpha \in R$. It is called the *Weyl group* of $\mathcal{R}$. Put

$$W^*(\mathcal{R}) = W(\mathcal{R}^*).$$

Then $W(\mathcal{R})$ acts on R , $\Gamma_0(R)$, $\mathcal{V}(R)$, M and V . If $w \in W(\mathcal{R})$ and $x \in M$ (resp. $x \in V$), one has $wx - x \in \Gamma_0(R)$ (resp. $wx - x \in \mathcal{V}(R)$); this is immediate from formula (1). Similarly for $W^*(\mathcal{R})$.

Lemma 1.1.9. *For all $\alpha \in R$, $x \in V$, $u \in V^*$, one has*

$$(4) \quad (s_\alpha^*(u), s_\alpha(x)) = (u, x).$$

Indeed, taking the scalar product of (1) and (1*), one finds that the left-hand side is equal to $(u, x) + (u, \alpha)(\alpha^*, x)((\alpha^*, \alpha) - 2) = (u, x)$.

Remark 1.1.10. If one supposes $0 \notin R$ and $0 \notin R^*$, then 1.1.9 is equivalent to (DR I).

Corollary 1.1.11. *Let $h \longmapsto h^\vee$ denote the isomorphism from $\text{GL}(M)$ onto $\text{GL}(M^*)$ associating with h its contragredient. Then formula (4) is also written*

$$(5) \quad (s_\alpha)^\vee = s_\alpha^*.$$

Corollary 1.1.12. *The preceding isomorphism induces an isomorphism from $W(\mathcal{R})$ onto $W^*(\mathcal{R})$.*

Scholium 1.1.13. *By the preceding result, we shall identify W and W^* , and consider W as a group of transformations of R , R^* , M , M^* , $\Gamma_0(R)$, $\Gamma_0(R^*)$, V , V^* , $\mathcal{V}(R)$, $\mathcal{V}(R^*)$. We shall write s_α for s_α^* .*

⁽²⁾**N.D.E.** One will note that $(\alpha^*)^* = \alpha$.

1.2. The map p

Lemma 1.2.1. *Let $p : M \longrightarrow M^*$ (resp. $V \longrightarrow V^*$) be the linear map defined by*

$$(6) \quad p(x) = \sum_{u \in R^*} (u, x)u.$$

Put $\ell(x) = (p(x), x)$. One has the following properties:

$$(7) \quad \ell(x) \geq 0, \quad \ell(\alpha) > 0 \quad \text{for } \alpha \in R.$$

$$(8) \quad \ell(wx) = \ell(x), \quad \text{for } w \in W.$$

$$(9) \quad (p(x), y) = (p(y), x), \quad \text{for } x, y \in V.$$

$$(10) \quad \ell(\alpha)\alpha^* = 2p(\alpha), \quad \text{for } \alpha \in R.$$

The first three relations are immediate.⁽³⁾ Let us prove the last. By (1), for $u \in V^*$, one has:

$$\begin{aligned} (u, \alpha)^2 \alpha^* &= (u, \alpha)u - (u, \alpha)s_\alpha(u) \\ &= (u, \alpha)u + (u, s_\alpha(\alpha))s_\alpha(u) \\ &= (u, \alpha)u + (s_\alpha(u), \alpha)s_\alpha(u) \quad (\text{since } s_\alpha^2 = \text{id}). \end{aligned}$$

Since s_α is a permutation of R^* (by (DR II)), one need only sum over $u \in R^*$ to conclude.

Scholium 1.2.2. *Relation (10) says that $\alpha \longmapsto \ell(\alpha)\alpha^*$ is the restriction to R of a linear map from M to M^* . In particular, one has*

$$(-\alpha)^* = -\alpha^*.$$

Corollary 1.2.3. *The map p induces an isomorphism from $\mathcal{V}(R)$ onto $\mathcal{V}(R)^*$.*

Indeed, p sends $\mathcal{V}(R)$ onto $\mathcal{V}(R^*)$. One therefore has

$$\dim(\mathcal{V}(R)) \geq \dim(\mathcal{V}(R^*)).$$

Applying this formula to the dual root datum, one deduces

$$\dim(\mathcal{V}(R)) = \dim(\mathcal{V}(R^*)),$$

hence p , being surjective, is also bijective.

Corollary 1.2.4. *One has $\dim(\mathcal{V}(R)) = \dim(\mathcal{V}(R^*))$, hence*

$$\text{rgss}(\mathcal{R}) = \text{rgss}(\mathcal{R}^*).$$

Corollary 1.2.5. *The bilinear form (u, x) is nondegenerate on $\mathcal{V}(R^*) \times \mathcal{V}(R)$, hence puts these $\mathbf{Q}$ -vector spaces in duality.*

Indeed, if $(u, x) = 0$ for every $u \in R^*$, then $p(x) = 0$.

Corollary 1.2.6. *The symmetric bilinear form $(p(x), y)$ is positive nondegenerate on $\mathcal{V}(R)$.*

Corollary 1.2.7. *W acts faithfully on R (and hence on the other sets of 1.1.13).*

Indeed, let $u \in V^*$. Let $w \in W$; suppose $w(\alpha) = \alpha$ for every $\alpha \in R$, and let us prove $w(u) = u$. One has

$$(w(u) - u, \alpha) = (w(u), \alpha) - (u, \alpha) = (u, w^{-1}(\alpha)) - (u, \alpha) = 0.$$

But $w(u) - u \in \mathcal{V}(R^*)$. If it is orthogonal to all the roots, it is zero by 1.2.5.

Corollary 1.2.8. *The group W is finite.*

⁽³⁾**N.D.E.** (8) follows from (4), (2) and (DR II).

Proposition 1.2.9. *The actions of W respect the correspondence between roots and coroots. In other words, for $\alpha \in R$ and $w \in W$, one has*

$$w(\alpha^*) = w(\alpha)^*.$$

It suffices to check this for $w = s_\beta$, $\beta \in R$, i.e. to check the formula

$$s_\beta(\alpha^*) = s_\beta(\alpha)^*.$$

Now $\ell(s_\beta(\alpha))s_\beta(\alpha)^*/2$ equals:

$$\begin{aligned} p(s_\beta(\alpha)) &= \sum_{u \in R^*} (u, s_\beta(\alpha))u = \sum_{u \in R^*} (s_\beta(u), \alpha)u \\ &= \sum_{u \in R^*} (u, \alpha)s_\beta(u) = s_\beta(p(\alpha)); \end{aligned}$$

since $\ell(s_\beta(\alpha)) = \ell(\alpha)$, one obtains $s_\beta(\alpha)^* = s_\beta(2p(\alpha)/\ell(\alpha)) = s_\beta(\alpha^*)$.

Corollary 1.2.10. *If $\alpha \in R$ and $w \in W$, one has*

$$ws_\alpha w^{-1} = s_{w(\alpha)}.$$

Indeed, $ws_\alpha w^{-1}(x) = x - (\alpha^*, w^{-1}(x))w(\alpha) = x - (w(\alpha^*), x)w(\alpha)$, and this last term equals, by 1.2.9:

$$x - (w(\alpha)^*, x)w(\alpha) = s_{w(\alpha)}(x).$$

Corollary 1.2.11. *Let $R' \subset R$. For (M, M^*, R', R'^*) to be a root datum, it is necessary and sufficient that $\alpha, \beta \in R'$ imply $s_\alpha(\beta) \in R'$.*

2. Relations between two roots

2.1. Proportional roots

Proposition 2.1.1. *Let α and β be two roots. The following conditions are equivalent:*

- (i) *There exists $k \in \mathbf{Q}$ such that $\alpha = k\beta$.*
- (ii) *$s_\alpha = s_\beta$.*

Moreover, under these conditions, one has $\alpha^* = k^{-1}\beta^*$, and k is equal to one of the numbers 1, -1, 2, -2, 1/2, -1/2.

Suppose first (i). First, one has

$$\alpha^* = \ell(\alpha)^{-1}2p(\alpha) = k^{-2}\ell(\beta)^{-1}2kp(\beta) = k^{-1}\beta^*.$$

This implies at once $s_\alpha = s_\beta$. Conversely, if one has $s_\alpha = s_\beta$, then

$$\alpha = s_\alpha(-\alpha) = s_\beta(-\alpha) = (\beta^*, \alpha)\beta - \alpha,$$

whence $2\alpha = (\beta^*, \alpha)\beta$, with $(\beta^*, \alpha) \in \mathbf{Z}$, hence (ii) implies (i). Finally, if $\alpha = k\beta$, then $\alpha^* = k^{-1}\beta^*$, whence

$$(\alpha^*, \beta) = 2k^{-1}, \quad (\beta^*, \alpha) = 2k,$$

hence $2k$ and $2k^{-1}$ are integers, and the proof is finished.

Application 2.1.2. Root data $\mathcal{R}$ such that $\text{rgss}(\mathcal{R}) = 1$ are of one of the following two types:

- (i) Type A_1 : there exists $\alpha \in M$ such that the roots are α and $-\alpha$. The coroots are then α^* and $-\alpha^*$.
- (ii) Type A'_1 : there exists $\alpha \in M$ such that the roots are $\alpha, -\alpha, 2\alpha, -2\alpha$. The coroots are then $\alpha^*, -\alpha^*, \alpha^*/2, -\alpha^*/2$.

Definition 2.1.3. One says that $\alpha \in R$ is *indivisible* if $\alpha/2 \notin R$. One says that $\mathcal{R}$ is *reduced* if every root is indivisible.

For $\mathcal{R}$ to be reduced, it is necessary and sufficient that $\mathcal{R}^*$ be so. If α is indivisible and $w \in W$, then $w(\alpha)$ is indivisible.

Definition 2.1.4. Let $\alpha \in R$. If α is indivisible, put $\text{ind}(\alpha) = \alpha$. Otherwise, put $\text{ind}(\alpha) = \alpha/2$.

Corollary 2.1.5. If $\alpha \in R$ is indivisible and $k\alpha \in R$, where $k \in \mathbf{Q}$, then $k \in \mathbf{Z}$.

Proposition 2.1.6. Let $\mathcal{R}$ be a root datum. Then

$$\text{ind}(\mathcal{R}) = (M, M^*, \text{ind}(R), \text{ind}(R)^*)$$

is a reduced root datum, and one has

$$W(\text{ind}(\mathcal{R})) = W(\mathcal{R}).$$

Indeed, $\text{ind}(\mathcal{R})$ is a root datum by 1.2.11, since the Weyl group permutes the indivisible roots. The second assertion follows from 2.1.1.

Remark 2.1.7. If $\mathcal{R}$ is not reduced, one has $\text{ind}(R)^* \neq \text{ind}(R^*)$, and hence $\text{ind}(\mathcal{R}^*) \neq \text{ind}(\mathcal{R})^*$.

2.2. Orthogonal roots

Lemma 2.2.1. Let α and β be two roots. One has

$$(11) \quad \ell(\alpha)(\alpha^*, \beta) = \ell(\beta)(\beta^*, \alpha).$$

This follows at once from 1.2.1, formulas (9) and (10).

Corollary 2.2.2. Let $\alpha, \beta \in R$. The following conditions are equivalent:

- (i) $(\alpha^*, \beta) = 0$, (i bis) $(\beta^*, \alpha) = 0$,
- (ii) $(p(\alpha), \beta) = 0$,
- (iii) $s_\alpha(\beta) = \beta$, (iii bis) $s_\beta(\alpha) = \alpha$,
- (iv) $s_\alpha(\beta^*) = \beta^*$, (iv bis) $s_\beta(\alpha^*) = \alpha^*$,
- (v) $s_\alpha \neq s_\beta$, and s_α and s_β commute.

All the equivalences are immediate except those involving (v). Let us show that (i) (and (i bis)) imply (v). One has

$$s_\alpha s_\beta(x) = x - (\beta^*, x)\beta - (\alpha^*, x)\alpha + (\beta^*, x)(\alpha^*, \beta)\alpha.$$

If $(\alpha^*, \beta) = 0$, then $(\beta^*, \alpha) = 0$, and $s_\alpha s_\beta(x) = s_\beta s_\alpha(x)$.

Conversely, suppose (v). One then has

$$s_\alpha = s_\beta s_\alpha s_\beta = s_{s_\beta(\alpha)} \quad (\text{by 1.2.10}).$$

By 2.1, there therefore exists $k \in \mathbf{Q}^*$ such that $\alpha = ks_\beta(\alpha) = k\alpha - k(\beta^*, \alpha)\beta$. Since $s_\alpha \neq s_\beta$, β and α are not proportional by 2.1.1, hence $(\beta^*, \alpha) = 0$.

Definition 2.2.3. Two roots satisfying the equivalent conditions of 2.2.2 are called *orthogonal*.

Remark 2.2.4. The roots α and β are orthogonal if and only if the coroots α^* and β^* are orthogonal.

Lemma 2.2.5. If α and β are two orthogonal roots, then $\alpha + \beta \in R$ if and only if $\alpha - \beta \in R$.

Indeed, $s_\beta(\alpha + \beta) = s_\beta(\alpha) + s_\beta(\beta) = \alpha - \beta$.

Lemma 2.2.6. *Let α and β be two nonorthogonal roots. If one defines $\ell(\gamma^*)$ for a coroot γ^* as ℓ of the root γ^* of $\mathcal{R}^*$, one has the relation*

$$(12) \quad \ell(\alpha)\ell(\alpha^*) = \ell(\beta)\ell(\beta^*).$$

Indeed, multiplying formula (11) by the corresponding formula for $\mathcal{R}^*$, and taking account of $(\gamma^*)^* = \gamma$ for every $\gamma \in R$, one finds:

$$(\beta^*, \alpha)(\alpha^*, \beta)\ell(\alpha)\ell(\alpha^*) = (\beta^*, \alpha)(\alpha^*, \beta)\ell(\beta)\ell(\beta^*).$$

2.3. General case

Proposition 2.3.1. *If α and β are any two roots, one has*

$$(13) \quad 0 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 4.$$

If α and β are neither proportional nor orthogonal, one has

$$1 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 3.$$

Indeed, one has $4(p(\alpha), \beta)^2 = \ell(\alpha)\ell(\beta)(\alpha^*, \beta)(\beta^*, \alpha)$. On the other hand, by 1.2.6, the symmetric bilinear form $(p(x), y)$ is positive nondegenerate on $\mathcal{V}(R)$, whence $(p(x), y)^2 \leq \ell(x)\ell(y)$.⁽⁴⁾

Corollary 2.3.2. *Let α and β be two nonorthogonal roots. If $\ell(\alpha) = \ell(\beta)$, there exists $w \in W$ such that $w(\alpha) = \beta$.*

Indeed, if α and β are proportional, since $\ell(\alpha) = \ell(\beta)$, one has $\alpha = \beta$ or $\alpha = -\beta$; in this case take $w = 1$ or $w = s_\alpha$. If α and β are neither proportional nor orthogonal, by formula (11) and 2.3.1 one has:

$$(\beta^*, \alpha) = (\alpha^*, \beta) = \pm 1.$$

If $(\beta^*, \alpha) = (\alpha^*, \beta) = 1$, take $w = s_\beta s_\alpha s_\beta$. If $(\beta^*, \alpha) = (\alpha^*, \beta) = -1$, take $w = s_\alpha s_\beta$.

Corollary 2.3.3. *If α and β are two roots, if $\alpha \neq \beta$ and $(\beta^*, \alpha) > 0$ (resp. if $\alpha \neq -\beta$ and $(\beta^*, \alpha) < 0$), then $\alpha - \beta$ (resp. $\alpha + \beta$) is a root.*

The second case is deduced from the first by changing β to $-\beta$. If α and β are proportional and $(\beta^*, \alpha) > 0$, one has $\beta = \alpha$, $2\beta = \alpha$, or $\beta = 2\alpha$. The first case is excluded. In the others, one respectively has $\alpha - \beta = \beta \in R$ or $\alpha - \beta = -\alpha \in R$. If α and β are not proportional, (α^*, β) and (β^*, α) are two strictly positive integers whose product is at most 3. One of them is therefore 1. If $(\beta^*, \alpha) = 1$, one has $\alpha - \beta = s_\beta(\alpha) \in R$; if $(\alpha^*, \beta) = 1$, one has $\alpha - \beta = -s_\alpha(\beta) \in R$.

Lemma 2.3.4. *Let α and β be two nonproportional roots. If $\beta - \alpha$ is not a root, then $\beta + k\alpha \in R$ for $k = -(\alpha^*, \beta)$, but not for $k = -(\alpha^*, \beta) + 1$.*

Indeed, one has $\beta - (\alpha^*, \beta)\alpha = s_\alpha(\beta) \in R$, but $\beta + (-(\alpha^*, \beta) + 1)\alpha = s_\alpha(\beta - \alpha) \notin R$.

Proposition 2.3.5. *Let α and β be two nonproportional roots. The set of integers k such that $\beta + k\alpha \in R$ is an interval $[p, q]$ ($p \leq 0$, $q \geq 0$), and one has $p + q = -(\alpha^*, \beta)$.*

For the first assertion, it suffices, for example, to prove that if $\beta + k\alpha \in R$, k an integer > 0 , then $\beta + (k - 1)\alpha \in R$. If $k = 1$, this is trivial. If $k \geq 2$, one has

$$(\alpha^*, \beta + k\alpha) = (\alpha^*, \beta) + 2k \geq -3 + 4 > 0,$$

and one concludes by 2.3.3. Let therefore $[p, q]$ be the interval in question. Applying 2.3.4 to $\beta + p\alpha$, one finds

$$q - p = -(\alpha^*, \beta + p\alpha) = -(\alpha^*, \beta) - 2p.$$

⁽⁴⁾**N.D.E.** With equality if and only if x and y are proportional.

(5)

Remark 2.3.6. The preceding formula contains the qualitative statements 2.2.5 and 2.3.3.

Supplements 2.3.7. ⁽⁶⁾By 1.2.1 (9) and 1.2.6, the bilinear form on $\mathcal{V}(R)$ defined by

$$\langle x, y \rangle = (p(x), y)$$

is symmetric and positive definite. By 1.2.1 (10), for every $\alpha \in R$ and $y \in \mathcal{V}(R)$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y),$$

where $\ell(\alpha) = \langle \alpha, \alpha \rangle$. Consequently, one deduces from 2.3.3 the following corollary.

Corollary. *Let $\alpha \neq \beta$ in R . If $\alpha - \beta \notin R$, then $\langle \alpha, \beta \rangle \leq 0$.*

3. Simple roots, positive roots

3.1. Systems of simple roots

Lemma 3.1.1. *Let α and α_i , $i = 1, \dots, n$, be roots. Suppose α distinct from the α_i . If one has a relation*

$$\alpha = \sum_{i=1}^n q_i \alpha_i, \quad q_i \in \mathbf{Q}_+,$$

there exists an index i such that $q_i \neq 0$, $(\alpha^, \alpha_i) > 0$, $\alpha - \alpha_i \in R$.*

Indeed, one writes $2 = (\alpha^*, \alpha) = \sum_{i=1}^n q_i (\alpha^*, \alpha_i)$, which proves the first two assertions. The third then follows from 2.3.3.

Proposition 3.1.2. *Let α and α_i , $i = 1, \dots, n$, be roots. If*

$$\alpha = \sum_{i=1}^n m_i \alpha_i, \quad m_i \in \mathbf{N}_+,$$

there exists a sequence $\beta_1, \dots, \beta_m$ of roots taken from among the α_i (not necessarily pairwise distinct) such that if one puts

$$\gamma_p = \sum_{i=1}^p \beta_i, \quad p = 1, \dots, m,$$

one has $\gamma_p \in R$ and $\gamma_m = \alpha$.

Let us argue by induction on the integer $\sum m_i = m'$. If α is equal to one of the α_i , say α_{i_0} (which is automatic if $m' = 1$), take $m = 1$, $\beta_1 = \alpha_{i_0}$. Otherwise, apply the preceding lemma, and there exists an index i such that $m_i \neq 0$ and $\alpha - \alpha_i$ is a root. One then has $(m_i - 1) \in \mathbf{N}$ and

$$\alpha - \alpha_i = (m_i - 1)\alpha_i + \sum_{j \neq i} m_j \alpha_j.$$

One need only apply the induction hypothesis to $\alpha - \alpha_i$.

Corollary 3.1.3. *Let $R' \subset R$. The following conditions are equivalent:*

- (i) *If $\alpha, \beta \in R'$ and $\alpha + \beta \in R$, then $\alpha + \beta \in R'$.*
- (ii) *$(\mathbf{N} \cdot R') \cap R \subset R'$.*

⁽⁵⁾**N.D.E.** $+2p$ has been corrected to $-2p$.

⁽⁶⁾**N.D.E.** These supplements, useful for the proof of 3.1.5, have been added.

Indeed, clearly (ii) $\Rightarrow$ (i). The converse follows at once from the proposition.

Definition 3.1.4. A set of roots satisfying the conditions of 3.1.3 is said to be *closed*.

Proposition 3.1.5. Let $\Delta \subset R$ be a set of roots. The following conditions are equivalent:

(i) The elements of Δ are indivisible, linearly independent, and

$$R \subset (\mathbf{Q}_+ \cdot \Delta) \cup (-\mathbf{Q}_+ \cdot \Delta).$$

(ii) The elements of Δ are linearly independent and

$$R \subset (\mathbf{N} \cdot \Delta) \cup (-\mathbf{N} \cdot \Delta).$$

(iii) Every root has a unique expression as a linear combination of the elements of Δ , with integer coefficients all of the same sign.

(7) Clearly (ii) $\Rightarrow$ (iii). Denote by $\alpha_1, \dots, \alpha_n$ the (distinct) elements of Δ . Let us prove (i) $\Rightarrow$ (ii). Let $\alpha \in R$. One therefore has a unique expression

$$\varepsilon\alpha = \sum q_i \alpha_i, \quad q_i \in \mathbf{Q}_+, \quad \varepsilon = \pm 1.$$

Let us show that the q_i are integers. This is certainly true if they are all zero except one (cf. 2.1.5). Otherwise, α is distinct from the α_i and, applying 3.1.1, one finds an i_0 such that $\alpha' = \varepsilon\alpha - \alpha_{i_0} \in R$ and $q_{i_0} \neq 0$. This gives

$$\alpha' = (q_{i_0} - 1)\alpha_{i_0} + \sum_{i \neq i_0} q_i \alpha_i.$$

Since at least one of the q_i , $i \neq i_0$, is nonzero, (i) implies $q_{i_0} - 1 \geq 0$. One repeats the operation for α' , and after finitely many steps one has proved that the q_i are integers.

Let us show (iii) $\Rightarrow$ (i). Let us show that $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly independent over $\mathbf{Q}$. Otherwise, one would have an equality

$$x = \sum_{i \in I} a_i \alpha_i = \sum_{j \in J} b_j \alpha_j,$$

where I, J are two disjoint subsets of $\{1, \dots, n\}$, at least one of them, say I , being nonempty, and $a_i, b_j \in \mathbf{N}^*$. By corollary 2.3.7, one has $\langle \alpha_i, \alpha_j \rangle \leq 0$ if $i \in I$ and $j \in J$, whence $\langle x, x \rangle \leq 0$, and hence $x = 0$. Let $i_0 \in I$; then the equalities

$$\alpha_{i_0} = \alpha_{i_0} + \sum_{i \in I} a_i \alpha_i = \alpha_{i_0} + \sum_{j \in J} b_j \alpha_j$$

imply (by (iii)) $a_i = 0 = b_j$ for all i, j , a contradiction. This shows that the elements of Δ are linearly independent over $\mathbf{Q}$; let us show that they are also indivisible.

Let therefore $\beta \in \Delta$ be divisible. One has $\beta/2 \in R$, whence

$$\frac{\beta}{2} = \varepsilon \sum_{\alpha \in \Delta} m_\alpha \alpha, \quad m_\alpha \in \mathbf{N}, \quad \varepsilon = \pm 1,$$

and hence also $\beta = 2\varepsilon \sum_{\alpha} m_\alpha \alpha$, whence, by uniqueness, $m_\alpha = 0$ if $\alpha \neq \beta$ and $2\varepsilon m_\beta = 1$, a contradiction.

Definition 3.1.6. A set Δ of roots satisfying the conditions of 3.1.5 is called a *system of simple roots*, or a *basis* of R .

If $w \in W$ and Δ is a system of simple roots, then $w(\Delta)$ is a system of simple roots.

Remark 3.1.7. This definition involves only R , and not $\mathcal{R}$; in fact, it even involves only $\text{ind}(R)$.

(7) **N.D.E.** In what follows, the order has been modified and the proof of the implication (iii) $\Rightarrow$ (i) expanded.

Remark 3.1.8. If Δ is a system of simple roots, Δ is a basis of the free abelian group $\Gamma_0(R)$. One therefore has $\text{Card}(\Delta) = \text{rgss}(\mathcal{R})$.

Remark 3.1.9. The conditions of 3.1.5 are then clearly equivalent to the following:

- (i') The elements of Δ are indivisible, number $\leq \text{rgss}(R)$, and $R \subset (\mathbf{Q}_+ \cdot \Delta) \cup (-\mathbf{Q}_+ \cdot \Delta)$.
- (ii') One has $\text{Card}(\Delta) \leq \text{rgss}(R)$, and $R \subset (\mathbf{N} \cdot \Delta) \cup (-\mathbf{N} \cdot \Delta)$.

Corollary 3.1.10. *If Δ is a system of simple roots, $\text{ind}(\Delta^*)$ is a system of simple coroots (i.e. a system of simple roots of R^*).*

Indeed, if $\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha$ ($a_\alpha \in \mathbf{N}$), then $p(\alpha) = \sum_{\alpha \in \Delta} a_\alpha p(\alpha)$, whence, by 1.2.1 (10):

$$\beta^* = \sum_{\alpha \in \Delta} a_\alpha \frac{\ell(\alpha)}{\ell(\beta)} \alpha^*,$$

which proves that $\text{ind}(\Delta^*)$ satisfies (i').

⁽⁸⁾By 2.1.1, if α^* is not indivisible, then $\alpha^* = 2u^*$, where $u^* \in \text{ind}(\Delta^*)$ (and $u = 2\alpha$). One deduces:

Corollary 3.1.11. *If Δ is a system of simple roots and $\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha$ ($a_\alpha \in \mathbf{Z}$) is the expression of β in Δ , then $2a_\alpha \ell(\alpha)$ is divisible by $\ell(\beta)$ (and even by $2\ell(\beta)$ if α^* is indivisible).*

Before continuing to state the properties of systems of simple roots, let us show that they exist.

3.2. Systems of positive roots

Definition 3.2.1. A set $R_+ \subset R$ is called a *system of positive roots* of $\mathcal{R}$ (or of R , cf. remark 3.2.2) if it satisfies the following conditions:

- (P 1) $R_+ \cap -R_+ = \emptyset$.
- (P 2) $R_+ \cup -R_+ = R$.
- (P 3) $(\mathbf{Q}_+ \cdot R_+) \cap R \subset R_+$.

In particular, such a set is closed. We shall see later (3.3.8) that in fact a closed set satisfying (P 1) and (P 2) also satisfies (P 3), hence is a system of positive roots. If $w \in W$ and R_+ is a system of positive roots, $w(R_+)$ is a system of positive roots.

Remark 3.2.2. This definition involves only R . One will also say that R_+ is a system of positive roots of R .

Remark 3.2.3. From (P 1) and (P 2), one deduces at once

$$\text{Card}(R_+) = \text{Card}(R)/2.$$

It follows that if R_+ and R'_+ are two systems of positive roots and $R_+ \subset R'_+$, then $R_+ = R'_+$.

Remark 3.2.4. If R_+ is a system of positive roots, R_+^* is a system of positive coroots (i.e. a system of positive roots of R^*).

This follows at once from 1.1.4 and 1.2.2.

Definition 3.2.5. Let Δ be a system of simple roots. Put

$$\mathcal{P}(\Delta) = (\mathbf{Q}_+ \cdot \Delta) \cap R = (\mathbf{N}_+ \cdot \Delta) \cap R.$$

⁽⁸⁾**N.D.E.** The following sentence has been added, and in 3.1.11 $4a_\alpha \ell(\alpha)$ has been replaced by $2a_\alpha \ell(\alpha)$.

Proposition 3.2.6. *If Δ is a system of simple roots, $\mathcal{P}(\Delta)$ is a system of positive roots. If Δ is a system of simple roots and R_+ a system of positive roots, one has the equivalence:*

$$\Delta \subset R_+ \iff R_+ = \mathcal{P}(\Delta).$$

The first assertion is immediate. If $\Delta \subset R_+$, then $\mathcal{P}(\Delta) \subset R_+$ by (P 3), hence $\mathcal{P}(\Delta) = R_+$ by 3.2.3. The rest is trivial.

Remark 3.2.7. There exist systems of positive roots: let $\geq$ be a totally ordered vector space structure on $\mathcal{V}(R)$. The set of roots ≥ 0 for this order relation is a system of positive roots.

Theorem 3.2.8. *Let R_+ be a system of positive roots. There exists a unique system of simple roots $\mathcal{S}(R_+)$ such that $\mathcal{S}(R_+) \subset R_+$, i.e. such that $R_+ = \mathcal{P}(\mathcal{S}(R_+))$.*

Uniqueness follows at once from:

Lemma 3.2.9. *Let Δ be a system of simple roots. Then for $\alpha \in \mathcal{P}(\Delta)$ to belong to Δ , it is necessary and sufficient that α not be the sum of two elements of $\mathcal{P}(\Delta)$.*

This lemma follows at once from the definitions and 3.1.2.

Let us now prove the existence of $\mathcal{S}(R_+)$. Consider the set of subsets T of R_+ such that $T = \text{ind}(T)$ and $(\mathbf{Q}_+ \cdot T) \supset R_+$. This set is nonempty, for it contains $\text{ind}(R_+)$. Let Δ be a minimal element of this set for inclusion. Let us show that Δ is a system of simple roots, i.e., by 3.1.5 (i), that Δ is a linearly independent subset of $M \otimes \mathbf{Q}$.

Lemma 3.2.10. *If $\alpha, \beta \in \Delta$ and $q\alpha + q'\beta \in R$, where $q, q' \in \mathbf{Q}$, then $qq' \geq 0$.*

Indeed, if $qq' < 0$, one can write (interchanging α and β if necessary)

$$a\alpha - b\beta \in R_+, \quad a, b \in \mathbf{Q}_+^*,$$

hence by hypothesis there exists a relation

$$a\alpha - b\beta = \sum_{\gamma \in \Delta} c(\gamma)\gamma, \quad c(\gamma) \in \mathbf{Q}_+.$$

If $a \leq c(\alpha)$, it is written

$$-\beta = \frac{c(\alpha) - a}{b}\alpha + \sum_{\gamma \neq \alpha} \frac{c(\gamma)}{b}\gamma.$$

Then this element belongs to $(\mathbf{Q}_+ \cdot \Delta) \cap R$, which is contained in R_+ by (P 3). But then $\beta \in R_+ \cap -R_+$, which contradicts (P 1).

If, on the contrary, $a > c(\alpha)$, one writes

$$(a - c(\alpha))\alpha = b\beta + \sum_{\gamma \neq \alpha} c(\gamma)\gamma,$$

which proves that $\Delta \subset \mathbf{Q}_+ \cdot (\Delta - \{\alpha\})$, contrary to the minimality of Δ . Recall (cf. 2.3.7) that the bilinear form on $\mathcal{V}(R)$ defined by $\langle x, y \rangle = (p(x), y)$ is a Euclidean scalar product; moreover, for $\alpha \in R$, one has $\langle \alpha, y \rangle = \ell(\alpha)(\alpha^*, y)/2$, so that $\langle \alpha, y \rangle$ and $\langle \alpha^*, y \rangle$ have the same sign.

Lemma 3.2.11. *If $\alpha, \beta \in \Delta$, then $\langle \beta^*, \alpha \rangle \leq 0$, and hence $\langle \beta, \alpha \rangle \leq 0$.*

Indeed, $s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta \in R$, whence $(\beta^*, \alpha) \leq 0$ by 3.2.10.

Let us now prove that Δ is linearly independent. Otherwise, 3.2.11 implies, as in the proof of 3.1.5, that there exists a nontrivial relation

$$\sum_{\alpha \in \Delta} m(\alpha)\alpha = 0, \quad m(\alpha) \in \mathbf{N},$$

whence $-\alpha_0 = (m(\alpha_0) - 1)\alpha_0 + \sum_{\alpha \neq \alpha_0} m(\alpha)\alpha$ if $m(\alpha_0) \geq 1$. Then, by (P 3), α_0 belongs to $R_+ \cap -R_+$, contradicting (P 1).

This shows that Δ is a basis of R and completes the proof of theorem 3.2.8.

Corollary 3.2.12. *Let R_+ be a system of positive roots, Δ a basis of R , and $w \in W$. One has:*

$$\begin{aligned}\Delta \subset R_+ &\iff R_+ = \mathcal{P}(\Delta) \iff \Delta = \mathcal{S}(R_+), \\ \mathcal{P}(\text{ind}(\Delta^*)) &= \mathcal{P}(\Delta)^*, \quad \mathcal{S}(R_+^*) = \text{ind}(\mathcal{S}(R_+)^*); \\ \mathcal{S}(w(R_+)) &= w(\mathcal{S}(R_+)), \quad \mathcal{P}(w(\Delta)) = w(\mathcal{P}(\Delta)).\end{aligned}$$

Definition 3.2.13. If one has chosen a system of simple roots Δ , the elements of $\mathcal{P}(\Delta)$ will be called *positive*. If one has chosen a system of positive roots R_+ , the elements of $\mathcal{S}(R_+)$ will be called *simple*.

Corollary 3.2.14. *Let R_+ be a system of positive roots. Let $\alpha \in R_+$. The following conditions are equivalent:*

- (i) α is simple (i.e. $\alpha \in \mathcal{S}(R_+)$).
- (ii) α is not the sum of two elements of R_+ .
- (iii) $R_+ - \{\alpha\}$ is closed (cf. 3.1.4).

The equivalence of (i) and (ii) follows at once from 3.2.9. The equivalence of (ii) and (iii) is immediate.

Definition 3.2.15. Let Δ be a system of simple roots. The sum of the coefficients of the decomposition of a root α in Δ is called the *order* of α relative to Δ , and is denoted $\text{ord}_\Delta(\alpha)$.

One has the equivalences:

$$\alpha \in \mathcal{P}(\Delta) \iff \text{ord}_\Delta(\alpha) > 0, \quad \alpha \in \Delta \iff \text{ord}_\Delta(\alpha) = 1.$$

Lemma 3.2.16. *Let Δ be a system of simple roots and $\alpha \in \mathcal{P}(\Delta)$. There exists a sequence $\alpha_i \in \Delta$, $i = 1, \dots, m$, such that*

$$\begin{aligned}\alpha_1 + \alpha_2 + \dots + \alpha_p &\in \mathcal{P}(\Delta) \quad \text{for } p = 1, \dots, m, \\ \alpha_1 + \alpha_2 + \dots + \alpha_m &= \alpha.\end{aligned}$$

Moreover, for every sequence α_i satisfying these conditions, one has $m = \text{ord}_\Delta(\alpha)$.

Trivial by 3.1.2.

3.3. Characterization and conjugacy of systems of positive roots

Lemma 3.3.1. *If $\alpha \in \mathcal{P}(\Delta)$, $\beta \in \Delta$, and α and β are not proportional, then $s_\beta(\alpha) \in \mathcal{P}(\Delta)$.*

Indeed, $s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta$. Since at least one simple root other than β occurs in the decomposition of α with a nonzero (hence strictly positive) coefficient, and therefore also in the decomposition of $s_\beta(\alpha)$ with the same coefficient, $s_\beta(\alpha)$ is also positive.

Corollary 3.3.2. *If $\beta \in \Delta$, the reflection s_β permutes the elements of $\mathcal{P}(\Delta)$ not proportional to β .*

Lemma 3.3.3. *If $\alpha \in \mathcal{P}(\Delta) - \Delta$, α indivisible, there exists $\beta \in \Delta$ such that $s_\beta(\alpha) \in \mathcal{P}(\Delta)$ and $\text{ord}_\Delta(s_\beta(\alpha)) < \text{ord}_\Delta(\alpha)$.*

Indeed, by 3.1.1 there exists $\beta \in \Delta$ such that $(\beta^*, \alpha) > 0$. Since $\alpha \notin \Delta$ and is indivisible, α cannot be proportional to β . Thus $s_\beta(\alpha) \in \mathcal{P}(\Delta)$ by 3.3.1, and one has $\text{ord}_\Delta(s_\beta(\alpha)) = \text{ord}_\Delta(\alpha) - (\beta^*, \alpha) < \text{ord}_\Delta(\alpha)$.

Corollary 3.3.4. *If $\alpha \in \mathcal{P}(\Delta)$ is indivisible, there exists a sequence $\beta_p \in \Delta$, $p = 1, \dots, q$, such that*

$$\alpha_p = s_{\beta_p} \cdots s_{\beta_1}(\alpha) \in \mathcal{P}(\Delta) \quad \text{for } p = 1, \dots, q,$$

and $\alpha_q \in \Delta$.

This follows from 3.3.3 by induction on $\text{ord}_\Delta(\alpha)$.

Proposition 3.3.5. *The Weyl group is generated by the s_α for $\alpha \in \Delta$. Every indivisible root is conjugate to a simple root by an element of the Weyl group.*

The second assertion follows at once from 3.3.4. The first follows from it by 1.2.10 and 2.1.1.

Proposition 3.3.6. *Let R_+ be a system of positive roots. Let $P' \subset R$ satisfy (P 2) and be closed. Then there exists $w \in W$ such that $w(R_+) \subset P'$.*

Let us state the corollaries immediately.

Corollary 3.3.7. *The Weyl group acts transitively on the set of systems of positive roots (resp. of systems of simple roots).*

Corollary 3.3.8. *For a subset of R to be a system of positive roots, it is necessary and sufficient that it satisfy (P 1) and (P 2) and be closed.*

Corollary 3.3.9. *If one equips $\Gamma_0(R)$ with an ordered group structure such that every root is > 0 or < 0 , the set of roots positive for this order structure is a system of positive roots.*

Let us now prove 3.3.6. One can find $w \in W$ such that $\text{Card}(w(R_+) \cap P')$ is maximal, hence, replacing R_+ by $w(R_+)$, one may suppose that

$$(*) \quad \text{Card}(R_+ \cap P') \geq \text{Card}(R_+ \cap s_\alpha(P'))$$

for every $\alpha \in \mathcal{S}(R_+) = \Delta$. Let us prove that $R_+ \subset P'$. Otherwise, since P' is closed, there exists $\alpha \in \Delta$, $\alpha \notin P'$. But since P' satisfies (P 2), one then has $-\alpha \in P'$ (hence $-2\alpha \in P'$ if 2α is a root). For every $\beta \in R_+ \cap P'$, one has $\beta \neq \alpha$; if 2α is not a root, one then has $s_\alpha(\beta) \in R_+$ (by 3.3.1), hence

$$s_\alpha(R_+ \cap P') \subset R_+ \cap s_\alpha(P');$$

but $R_+ \cap s_\alpha(P')$ also contains $\alpha = s_\alpha(-\alpha)$, which contradicts inequality (*). If 2α is a root, one reasons similarly.

To study the sets of roots satisfying (P 2) and closed, one is therefore reduced to the case where they contain a set of positive roots.

Proposition 3.3.10. *Let R_+ be a system of positive roots and P' a closed subset of R containing R_+ . If one puts $\Delta = \mathcal{S}(R_+)$ and $\Delta' = \Delta \cap -P'$, then P' is the union of R_+ and the set of roots which are linear combinations with negative coefficients of the elements of Δ' .*

Let us prove the assertion by induction on the order of a root $\gamma \in P' - R_+$. If $\text{ord}_\Delta(\gamma) = -1$, then $-\gamma \in \Delta$ and $\gamma \in -\Delta'$. If $\text{ord}_\Delta(\gamma) < -1$, there exists, by 3.1.2, $\beta \in \Delta$ such that $-\gamma - \beta \in R$; then

$$0 < \text{ord}_\Delta(-\gamma - \beta) = \text{ord}_\Delta(-\gamma) - 1 < \text{ord}_\Delta(-\gamma),$$

and the first inequality shows that $-\gamma - \beta \in R_+$. Thus, since $R_+ \cap -R_+ = \emptyset$, and since $\gamma + \beta$ is the sum of two roots of P' , it is an element of $P' - R_+$ such that $\text{ord}_\Delta(\gamma + \beta) > \text{ord}_\Delta(\gamma)$. Hence, by the induction hypothesis, $\gamma + \beta$ is a linear combination with negative coefficients of the elements of Δ' . Since $\gamma = (\gamma + \beta) - \beta$, it suffices to check that $\beta \in -P'$. Now $\gamma \in P'$ and $-(\gamma + \beta) \in R_+ \subset P'$, so that $-\beta = \gamma - (\gamma + \beta)$ belongs to P' .

Definition 3.3.10.1. ⁽⁹⁾One says that a subset R' of R is *symmetric* if $-R' = R'$.

⁽⁹⁾**N.D.E.** This definition has been inserted here.

Scholium 3.3.11. *Let P' be a set of roots satisfying (P 2) and closed. There exist a system of simple roots Δ and a subset Δ' of Δ such that*

$$P' = R \cap (\mathbf{N} \cdot \Delta \cup -\mathbf{N} \cdot \Delta').$$

If one puts $R_{\Delta'} = (\mathbf{Z} \cdot \Delta') \cap R$, which is a closed symmetric subset of R , then P' is therefore the disjoint union of $R_{\Delta'}$ and the closed subset of $\mathcal{P}(\Delta)$ formed by the $\alpha \in \mathcal{P}(\Delta) - \mathbf{N} \cdot \Delta'$, i.e. the positive roots which in their decomposition in Δ “contain at least one root of $\Delta - \Delta'$ ”.

3.4. Closed symmetric sets of roots

Proposition 3.4.1. *Let $\mathcal{R} = (M, M^*, R, R'^*)$ be a root datum and R' a closed symmetric subset of R . Then:*

- (i) $\mathcal{R}' = (M, M^*, R', R'^*)$ is a root datum;
- (ii) for every system of positive roots R_+ of R , $R'_+ = R_+ \cap R'$ is a system of positive roots of R' ;
- (iii) the Weyl group $W(\mathcal{R}')$ of $\mathcal{R}'$ is the subgroup of $W(\mathcal{R})$ generated by the s_α for $\alpha \in R'$.

The first assertion is trivial by 1.2.11, the second follows from 3.3.8, and the third is obvious.

Corollary 3.4.2. *Let $w \in W(\mathcal{R}')$. The order of w is the least integer $n > 0$ such that $w^n(\alpha') = \alpha'$ for every $\alpha' \in R'$.*

It suffices to apply 1.2.7 to the root datum $\mathcal{R}'$.

Corollary 3.4.3. *Let α and β be two nonproportional roots. Let n be the least integer > 0 such that $(s_\alpha s_\beta)^n(\alpha) = \alpha$ and $(s_\alpha s_\beta)^n(\beta) = \beta$. Then the subgroup $W_{\alpha, \beta}$ of W generated by s_α and s_β is defined by the relations:*

$$s_\alpha^2 = 1, \quad s_\beta^2 = 1, \quad (s_\alpha s_\beta)^n = 1.$$

Taking account of 3.4.2, it suffices to check:

Lemma 3.4.4. *Let G be the group generated by two elements x and y subject to $x^2 = y^2 = 1$. Every invariant subgroup of G containing neither x nor y is generated (as an invariant subgroup) by an element of the form $(xy)^n$, where $n > 0$.*

⁽¹⁰⁾Indeed, every element of G is written $(xy)^n$, or $(yx)^n = (xy)^{-n}$, or:

$$\begin{aligned} (a) \quad & x(yx)^{2n} \quad \text{or} \quad y(xy)^{2n+1}, \\ (b) \quad & y(xy)^{2n} \quad \text{or} \quad x(yx)^{2n+1}, \end{aligned}$$

where $n \in \mathbf{N}$. Now the elements of type (a), resp. (b), are conjugate to x , resp. to y .

Remark 3.4.5. One immediately calculates the integer n : if one puts

$$(\alpha^*, \beta) = p, \quad (\beta^*, \alpha) = q,$$

one has⁽¹¹⁾

$$(s_\alpha s_\beta)(\alpha) = (pq - 1)\alpha - q\beta, \quad (s_\alpha s_\beta)(\beta) = p\alpha - \beta.$$

The integer n is therefore the order of the matrix

$$\begin{pmatrix} pq - 1 & p \\ -q & -1 \end{pmatrix}.$$

If $pq = 0$, then $p = q = 0$ by 2.2.2, whence $n = 2$. Otherwise, by 2.3.1, pq is 1, 2 or 3, and one finds, respectively, $n = 3, 4$ or 6.

⁽¹⁰⁾**N.D.E.** The original has been corrected in what follows.

⁽¹¹⁾**N.D.E.** The original has been corrected in what follows.

N. B. Writing that the order of the preceding matrix is finite, one recovers inequality (13) of 2.3.1.

Definition 3.4.6. Let Δ be a system of simple roots and $\Delta' \subset \Delta$. Put

$$R_{\Delta'} = R \cap (\mathbf{Q} \cdot S') = R \cap (\mathbf{Z} \cdot \Delta').$$

Lemma 3.4.7. $R_{\Delta'}$ is closed and symmetric, Δ' is a system of simple roots of the root datum $(M, M^*, R_{\Delta'}, R_{\Delta'}^*)$, whose Weyl group is the subgroup $W_{\Delta'}$ of W generated by the s_α for $\alpha \in \Delta'$. One has $\Delta \cap R_{\Delta'} = \Delta'$.

Trivial.

Proposition 3.4.8. Let $R' \subset R$. The following conditions are equivalent:

- (i) There exists a vector subspace V' of V (or of $\mathcal{V}(R)$) such that $R' = R \cap V'$.
- (ii) There exist a system of simple roots Δ of R and a subset Δ' of Δ such that $R' = R_{\Delta'}$.

More precisely, under these conditions, every system of simple roots Δ' of (M, M^*, R', R'^*) is contained in a system of simple roots Δ of R , and one has $R' = R_{\Delta'}$.

Clearly (ii) $\Rightarrow$ (i). Suppose (i) holds: then R' is closed and symmetric, hence

$$(M, M^*, R', R'^*)$$

is a root datum. Let Δ' be a system of simple roots of this datum and $R'_+ = \mathcal{P}(\Delta')$. If $V' = V$, then Δ' is a system of simple roots of R , and the proof is finished. Otherwise, there exists $x \in V^*$ such that

$$(x, V') = \{0\}, \quad (x, \alpha) \neq 0 \quad \text{for every } \alpha \in R - R'.$$

Put $R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}$ and $R_+ = R_+(x) \cup R'_+$. For every $\alpha \in R$, one has the equivalences

$$\begin{aligned} (x, \alpha) > 0 &\iff \alpha \in R_+(x), \\ (x, \alpha) < 0 &\iff \alpha \in -R_+(x), \\ (x, \alpha) = 0 &\iff \alpha \in R'. \end{aligned}$$

It follows at once from 3.3.8 that R_+ is a system of positive roots of R . Put $\Delta = \mathcal{S}(R_+)$. Clearly, it suffices to prove $\Delta' \subset \Delta$. Otherwise, let $\alpha \in \Delta' - \Delta$. Then, by 3.2.14, there exist $\beta, \gamma \in R_+$ such that $\alpha = \beta + \gamma$. If $\beta, \gamma \in R_+(x)$, one has $\alpha \in R_+(x)$, which is absurd since $(x, S') = 0$. If β or γ , for example β , belongs to R'_+ , then, since R' is symmetric and closed, $\gamma = \alpha - \beta$ belongs to $R_+ \cap R' = R'_+$; but then α is not simple in R'_+ .

Lemma 3.4.9. Under the preceding conditions, let $\alpha \in \mathcal{P}(\Delta) - R'$. For every $w \in W_{\Delta'}$, one has $w(\alpha) \in \mathcal{P}(\Delta) - R'$.

Indeed, it suffices to check this for $w = s_\beta$, $\beta \in \Delta'$, in which case it follows from 3.3.1 and $s_\beta(R') = R'$.

Lemma 3.4.10. Let $w \in W$. Under the conditions of 3.4.8, the following conditions are equivalent:

- (i) $w \in W_{\Delta'}$.
- (ii) For every $m \in M$, $w(m) - m \in V'$.
- (iii) For every $\alpha \in R$, $w(\alpha) - \alpha \in V'$.

Clearly (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). Let us prove (iii) $\Rightarrow$ (i). Let therefore $w \in W$ be such that $w(\alpha) - \alpha \in V'$ for every $\alpha \in R$. Write $w = s_{\alpha_n} \cdots s_{\alpha_1}$, with $\alpha_i \in \Delta$, and let us show by induction on n that each $\alpha_i \in \Delta'$. One may suppose that $w' = s_{\alpha_{n-1}} \cdots s_{\alpha_1} \in W_{\Delta'}$. One then has, for every $\alpha \in R$,

$$w(\alpha) - \alpha = w'(\alpha) - \alpha - (\alpha_n^*, w'(\alpha))\alpha_n.$$

Taking $\alpha = w'^{-1}(\alpha_n)$, one finds $2\alpha_n \in V'$, whence $\alpha_n \in \Delta'$, hence $w = s_{\alpha_n} w'$ belongs to $W_{\Delta'}$.

3.5. Various remarks

Proposition 3.5.1. *Let R_+ be a system of positive roots. Put*

$$\rho_{R_+} = \frac{1}{2} \sum_{\alpha \in \text{ind}(R_+)} \alpha.$$

Then $(\beta^, \rho_{R_+}) = 1$ for every $\beta \in \mathcal{S}(R_+)$.*

Indeed, one can write

$$2\rho_{R_+} = \beta + \sum_{\substack{\alpha \in \text{ind}(R_+) \\ \alpha \neq \beta}} \alpha,$$

hence $s_\beta(2\rho_{R_+}) = 2\rho_{R_+} - 2\beta$, by 3.3.2.

Corollary 3.5.2. *Put $\rho_{R_+}^* = \frac{1}{2} \sum_{\alpha \in \text{ind}(R_+^*)} \alpha^*$. Then:*

- (i) $(\rho_{R_+}^*, \alpha) > 0$ for every $\alpha \in R_+$ (i.e. $\rho_{R_+}^* \in \mathcal{C}(R_+)$, cf. 3.6.8).
- (ii) For every $\alpha \in R$, one has $\text{ord}_{\mathcal{S}(R_+)}(\alpha) = (\rho_{R_+}^*, \alpha)$.⁽¹²⁾

Remark 3.5.3. If $w \in W$, one has $\rho_{w(R_+)} = w(\rho_{R_+})$ and $\rho_{w(R_+)}^* = w(\rho_{R_+}^*)$.

Proposition 3.5.4. *Let α and γ be two nonproportional roots, α being supposed indivisible. There exists a system of simple roots containing α and a root β such that $\gamma = a\alpha + b\beta$, with $a, b \in \mathbf{N}$.*

Indeed, construct a basis of the vector space $\mathcal{V}(R)$ containing $\alpha_1 = \alpha$, $\alpha_2 = \gamma$. Consider the lexicographic order relative to this basis. Denoting the set of roots > 0 by R_+ , it is clear that R_+ is a system of positive roots, and that⁽¹³⁾ the least element of R_+ not proportional to α is simple. This element has the form

$$\beta = p\alpha + q\gamma, \quad 0 < q \leq 1.$$

One therefore has $\gamma = q^{-1}\beta - q^{-1}p\alpha$, and, since $q^{-1} > 0$, one has $q^{-1} \in \mathbf{N}^*$ and $-q^{-1}p \in \mathbf{N}$.

Finally, let us make two remarks on the group $\Gamma_0(R)$.

Proposition 3.5.5. *Let G be an abelian group and $f : R \rightarrow G$ a map satisfying the following two conditions:*

- (i) If $\alpha \in R$, $f(-\alpha) = -f(\alpha)$.
- (ii) If $\alpha, \beta, \alpha + \beta \in R$, $f(\alpha + \beta) = f(\alpha) + f(\beta)$.

Then there exists a unique group homomorphism $\bar{f} : \Gamma_0(R) \rightarrow G$ such that $\bar{f}(\alpha) = f(\alpha)$ for $\alpha \in R$.

Indeed, if Δ is a system of simple roots of R , and $\beta \in R$ is written $\sum_{\alpha \in \Delta} a(\alpha)\alpha$, it follows at once from 3.2.16 that $f(\beta) = \sum_{\alpha \in \Delta} a(\alpha)f(\alpha)$. Now Δ is a basis of $\Gamma_0(R)$.

Proposition 3.5.6. *Let Δ be a system of simple roots. There exists on $\Gamma_0(R)$ a totally ordered group structure such that the roots > 0 are the elements of $\mathcal{P}(\Delta)$ and $\alpha \mapsto \text{ord}_\Delta(\alpha)$ is an increasing function.*

Indeed, let $\alpha_1, \alpha_2, \dots, \alpha_n$ ($n = \text{rgss}(\mathcal{R})$) be the elements of Δ . For $x \in \Gamma_0(R)$, one has a decomposition

$$x = \sum_{i=1}^n m_i(x)\alpha_i.$$

It suffices to take the lexicographic order relative to the functions $\sum m_i, m_n, m_{n-1}, \dots, m_2$.

⁽¹²⁾**N.D.E.** Indeed, by 3.1.5, it suffices to check the formula for $\alpha \in \Delta$. Now, by 3.5.1 applied to R^* , one has in this case $(\rho_{R_+}^*, \alpha) = 1 = \text{ord}_{\mathcal{S}(R_+)}(\alpha)$.

⁽¹³⁾**N.D.E.** The original has been modified to take account of the case where 2α is a root.

Remark 3.5.7. The first roots are, in order:

$$\alpha_1, \alpha_2, \dots, \alpha_n;$$

then one has (if they are roots) $2\alpha_1, \alpha_1 + \alpha_2, \alpha_1 + \alpha_3, \dots$

3.6. Weyl chambers

Lemma 3.6.1. *Let V be a finite-dimensional $\mathbf{R}$ -vector space.⁽¹⁴⁾ Let f_i be independent linear forms. Put*

$$C = \{x \in V \mid f_i(x) > 0\}.$$

Then C is a maximal convex subset of $X = V - \bigcup_i f_i^{-1}(0)$.

Trivial.

Definition 3.6.2. A subset C of V which can be described by the procedure of 3.6.1 will be called (here) a *chamber* of V .

Definition 3.6.3. One says that the hyperplane H of V is a *wall* of C if $H \cap (\overline{C} - C)$ contains a nonempty open subset of H .

Remark 3.6.4. For a convex subset, the closure can be described without appealing to the topology of V : it is the set of endpoints of all open segments contained in the given subset.

Lemma 3.6.5. *Under the conditions of 3.6.1, one has*

$$\overline{C} = \{x \in V \mid f_i(x) \geq 0\}.$$

The walls of C are the hyperplanes $f_i^{-1}(0)$.

This is clear for the first assertion. The second then follows from $\overline{C} - C \subset \bigcup_i f_i^{-1}(0)$ and the fact that the $f_i^{-1}(0)$ are clearly walls of C .

Proposition 3.6.6. *Let C be a chamber of V . If $H_i, i = 1, 2, \dots, n$, are the distinct walls of C , then for every system of linear forms $\{u_i\}$ such that $H_i = u_i^{-1}(0)$, there exist $\varepsilon_i \in \{-1, +1\}$ such that C is defined by*

$$C = \{x \in V \mid \varepsilon_i u_i(x) > 0\}.$$

For every wall H of C , one has $H \cap C = \emptyset$ and $H \cap \overline{C} = \overline{C_H}$, where C_H is a chamber in H . The walls of C_{H_i} are the $H_j \cap H_i$ for $j \neq i$.

This follows trivially from the lemma.

Definition 3.6.7. The C_{H_i} are the *faces* of C .

Let now $\mathcal{R} = (M, M^*, R, R^*)$ be a root datum. Put $V_{\mathbf{R}}^* = M^* \otimes \mathbf{R}$.

Definition 3.6.8. ⁽¹⁵⁾For every $\alpha \in R$, put

$$\mathcal{H}_\alpha = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) = 0\}.$$

Put $X = V_{\mathbf{R}}^* - \bigcup_{\alpha \in R} \mathcal{H}_\alpha$. For every $x \in X$, put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}.$$

For every system of positive roots R_+ , put

$$\mathcal{C}(R_+) = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in R_+\}.$$

Proposition 3.6.9.

⁽¹⁴⁾**N.D.E.** $\mathbf{Q}$ has been replaced by $\mathbf{R}$.

⁽¹⁵⁾**N.D.E.** The definition of the hyperplanes $\mathcal{H}_\alpha$ has been added.

(i) For every $x \in X$, $R_+(x)$ is a system of positive roots. For every system of positive roots R_+ , $\mathcal{C}(R_+)$ is a chamber in $V_{\mathbf{R}}^*$. The $\mathcal{C}(R_+)$ are the maximal convex subsets of X .

(ii) Let Δ be a system of simple roots. One has

$$\mathcal{C}(\mathcal{P}(\Delta)) = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in \Delta\}.$$

The walls of $\mathcal{C}(\mathcal{P}(\Delta))$ are the hyperplanes $\mathcal{H}_\alpha = \alpha^{-1}(0)$, for $\alpha \in \Delta$; its faces are the

$$C_\alpha = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) = 0, (x, \beta) > 0 \text{ for } \beta \in \Delta, \beta \neq \alpha\}.$$

(iii) One has the equivalence

$$R_+(x) = R_+ \iff x \in \mathcal{C}(R_+).$$

First, it is clear that $R_+(x)$ is a system of positive roots. Since $x \in \mathcal{C}(R_+(x))$, the union of the $\mathcal{C}(R_+(x))$ is X . Property (iii) is immediate; it follows that the $\mathcal{C}(R_+)$ form a partition of X . The first assertion of (ii) is obvious. It follows at once that $\mathcal{C}(R_+)$ is a chamber in V , which proves the rest of (i). Putting $C = \mathcal{C}(\mathcal{P}(\Delta))$, one need only note that

$$\overline{C} \bigcup_{\alpha \in R} \alpha^{-1}(0) = \overline{C} \bigcup_{\alpha \in \Delta} \alpha^{-1}(0)$$

to complete the proof of (ii) by 3.6.1.

Definition 3.6.10. The $\mathcal{C}(R_+)$ are called the *Weyl chambers* of the root datum. On the other hand, for every Weyl chamber C , put $R_+(C) = R_+(x)$ for any $x \in C$.

Corollary 3.6.11. The maps $R_+ \mapsto \mathcal{C}(R_+)$ and $C \mapsto R_+(C)$ give a bijective correspondence between systems of positive roots and Weyl chambers.

This correspondence is invariant under the Weyl group:

Lemma 3.6.12. If $w \in W(\mathcal{R})$, one has $\mathcal{C}(w(R_+)) = w(\mathcal{C}(R_+))$.

Corollary 3.6.13. The correspondences $\Delta \leftrightarrow R_+ \leftrightarrow C$ are isomorphisms of homogeneous spaces under $W(\mathcal{R})$.

We shall see later that these homogeneous spaces are principal (5.5).

Remark 3.6.14. If C is a Weyl chamber, $-C$ is also one, called the *opposite* of C . There therefore exists $w_0 \in W$ (and in fact only one, cf. 5.5) such that $w_0(C) = -C$; it is called the *reflection* of the root datum relative to the Weyl chamber C (or relative to $R_+(C)$ or $\mathcal{S}(R_+(C)), \dots$).

4. Reduced root data of semisimple rank 2

4.0. — ⁽¹⁶⁾Let $\mathcal{R}$ be a root datum of semisimple rank 2. Let $\{\alpha, \beta\}$ be a system of simple roots. Suppose $\ell(\alpha) \leq \ell(\beta)$. By 2.3.1 and 3.2.1, one then has four possibilities:

Type	$\ell(\beta)/\ell(\alpha)$	$\ell(\beta^*)/\ell(\alpha^*)$	(β^*, α)	(α^*, β)
$A_1 \times A_1$	—	—	0	0
A_2	1	1	-1	-1
B_2	2	1/2	-1	-2
G_2	3	1/3	-1	-3

It then follows from 3.4.5 that the order of $s_\alpha s_\beta$ is respectively 2, 3, 4, 6.

Let us study each of these systems separately and give the list of indivisible roots.

Type $A_1 \times A_1$. The indivisible roots are $\alpha, \beta, -\alpha, -\beta$. The corresponding coroots are $\alpha^*, \beta^*, -\alpha^*, -\beta^*$.

⁽¹⁶⁾**N.D.E.** The numbering 4.0 has been added for subsequent references.

Type A_2 . The positive indivisible roots are as follows:

root γ	α	β	$\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	1	1
coroot γ^*	α^*	β^*	$\alpha^* + \beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	1	1	1

The half-sum of the positive indivisible roots is $\rho = \alpha + \beta$.

Type B_2 . The positive indivisible roots are the following:

root γ	α	β	$\alpha + \beta$	$2\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	2	1	2
coroot γ^*	α^*	β^*	$2\alpha^* + \beta^*$	$\alpha^* + \beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	2	1	2	1

The half-sum of the positive indivisible roots is $\rho = (4\alpha + 3\beta)/2$.

Type G_2 . The positive indivisible roots are the following:

root γ	α	β	$\alpha + \beta$	$2\alpha + \beta$	$3\alpha + \beta$	$3\alpha + 2\beta$
$\ell(\gamma)/\ell(\alpha)$	1	3	1	1	3	3
coroot γ^*	α^*	β^*	$\alpha^* + 3\beta^*$	$2\alpha^* + 3\beta^*$	$\alpha^* + \beta^*$	$\alpha^* + 2\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	3	1	3	3	1	1

The half-sum of the positive indivisible roots is $\rho = 5\alpha + 3\beta$.

Proposition 4.1. ^(*)Let n be the order of $s_\alpha s_\beta$. Put $u_0 = 0$ and, for $p \in \mathbf{N}$,

$$\begin{cases} u_{2p+1} = u_{2p} + (s_\alpha s_\beta)^p(\alpha); \\ u_{2p+2} = u_{2p+1} + (s_\alpha s_\beta)^p s_\alpha(\beta). \end{cases}$$

Then:⁽¹⁷⁾

- (i) $u_{k+2n} = u_k$ for every $k \in \mathbf{N}$.
- (ii) $u_0 = 0$, $u_1 = \alpha$, $u_{2n-1} = \beta$, $u_{2n} = 0$.
- (iii) If $1 < k < 2n - 1$, one has $u_k = a_k \alpha + b_k \beta$ with $a_k, b_k \in \mathbf{N}^*$.

Assertion (i) follows from $(s_\alpha s_\beta)^n = 1$ and $u_{2n} = 0$.

Let us prove (ii) and (iii). Calculation gives at once, in the four cases, the sequences of values:⁽¹⁸⁾

$$\begin{aligned} (A_1 \times A_1) & 0, \alpha, \beta + \alpha, \beta, 0. \\ (A_2) & 0, \alpha, \beta + 2\alpha, 2\beta + 2\alpha, 2\beta + \alpha, \beta, 0. \\ (B_2) & 0, \alpha, \beta + 3\alpha, 2\beta + 4\alpha, 3\beta + 4\alpha, 3\beta + 3\alpha, 2\beta + \alpha, \beta, 0. \\ (G_2) & 0, \alpha, \beta + 4\alpha, 2\beta + 6\alpha, 4\beta + 9\alpha, 5\beta + 10\alpha, 6\beta + 10\alpha, \\ & 6\beta + 9\alpha, 5\beta + 6\alpha, 4\beta + 4\alpha, 2\beta + \alpha, \beta, 0. \end{aligned}$$

Lemma 4.2. Put $w_{2p} = (s_\beta s_\alpha)^p$, $w_{2p+1} = s_\alpha (s_\beta s_\alpha)^p$, so that $w_0, \dots, w_{2n-1}$ are the distinct elements of W . Let $u_0, \dots, u_{2n-1}$ be the elements of V defined in 4.1. For every $x \in V^*$, one has

$$x - w_k(x) = n_k \alpha^* + m_k \beta^*,$$

with $n_k, m_k \in \mathbf{Q}$ and $n_k + m_k = (x, u_k)$.⁽¹⁹⁾

^(*)The following paragraphs 4.1 to 4.4 are used in the proof of 5.1. There are now simpler proofs of 5.1 (see [BLie], § V.3, Th. 1). **N.D.E.** The reference has been specified, and this Note, which in the original appeared in 4.2, has been placed here.

⁽¹⁷⁾**N.D.E.** Let ρ be the half-sum of the positive roots (cf. 3.5.1); if one puts, as in 4.2 below, $w_0 = 1$, $w_1 = s_\alpha$, $w_2 = s_\beta s_\alpha$, $w_3 = s_\alpha s_\beta s_\alpha$, etc., then u_k is precisely $\rho - w_k^{-1}(\rho)$, which proves (i), since $w_{2n} = \text{id}$.

⁽¹⁸⁾**N.D.E.** The original has been corrected to agree with the convention $\ell(\alpha) \leq \ell(\beta)$ of 4.0.

⁽¹⁹⁾**N.D.E.** Taking account of N.D.E. (17), this follows immediately from 3.5.1 and 1.1.9: one has $n_k + m_k = (x - w_k(x), \rho) = (x, \rho - w_k^{-1}(\rho)) = (x, u_k)$.

The proof is by induction on k . If $k = 0$, the formula is trivially satisfied. Let us carry out, for example, the step from w_{2p} to w_{2p+1} . One has $w_{2p+1} = s_\alpha w_{2p}$, whence

$$\begin{aligned} x - w_{2p+1}(x) &= x - w_{2p}(x) + w_{2p}(x) - s_\alpha w_{2p}(x) \\ &= n_{2p}\alpha^* + m_{2p}\beta^* + (w_{2p}(x), \alpha)\alpha^*. \end{aligned}$$

Thus

$$\begin{aligned} n_{2p+1} + m_{2p+1} &= n_{2p} + m_{2p} + ((s_\beta s_\alpha)^p(x), \alpha) \\ &= (x, u_{2p}) + (x, (s_\alpha s_\beta)^p(\alpha)) = (x, u_{2p+1}). \end{aligned}$$

Corollary 4.3. ⁽²⁰⁾ *Let $x \in V^*$. For every $w \in W$, put*

$$x - w(x) = a_w\alpha^* + b_w\beta^*.$$

If $(x, \alpha) \geq 0$ and $(x, \beta) \geq 0$, then $a_w + b_w \geq 0$. If in addition $(x, \alpha) > 0$ (resp. $(x, \beta) > 0$), then $a_w + b_w > 0$ for $w \neq 1, s_\beta$ (resp. for $w \neq 1, s_\alpha$).

This follows at once from 4.1 and 4.2.

Corollary 4.4. *Let $\mathcal{R}$ be any root datum and Δ a system of simple roots. Let γ be a positive root, α, β two simple roots, $W_{\alpha, \beta}$ the subgroup of W generated by s_α and s_β . If*

$$\text{ord}_\Delta(s_\alpha(\gamma)) < \text{ord}_\Delta(\gamma), \quad \text{ord}_\Delta(s_\beta(\gamma)) \leq \text{ord}_\Delta(\gamma),$$

then, for every $w \in W_{\alpha, \beta}$, one has $\text{ord}_\Delta(w(\gamma)) \leq \text{ord}_\Delta(\gamma)$; moreover, $\text{ord}_\Delta(w(\gamma)) < \text{ord}_\Delta(\gamma)$ if $w \neq 1, w \neq s_\beta$.

Indeed, consider the dual root datum $\mathcal{R}^*$, and then the datum (M^*, M, R'^*, R') , where R' is the set of roots which are rational linear combinations of α and β . Applying 4.3 to this datum, one obtains the announced corollary. ⁽²¹⁾

5. The Weyl group: generators and relations

Let $\mathcal{R}$ be a root datum. Since the Weyl group is the same for this datum and for the corresponding reduced datum, one may suppose $\mathcal{R}$ *reduced* to study the Weyl group.

Let $\Delta = \{\alpha_1, \dots, \alpha_n\}$ be a system of simple roots ($n = \text{rgss}(\mathcal{R})$). Let n_{ij} be the order of the element $s_{\alpha_i} s_{\alpha_j}$ of W . In particular, one has $n_{ii} = 1$, and we saw in 3.4.2 and 3.4.3 that the subgroup W_{ij} of W generated by s_{α_i} and s_{α_j} was defined by the relations:

$$s_{\alpha_i}^2 = s_{\alpha_j}^2 = (s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1.$$

Theorem 5.1. *The group W is the group generated by the elements s_{α_i} , $i = 1, 2, \dots, n$, subject to the relations $(s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1$.*

We have already seen that the theorem is true when $n = 2$; we shall use this remark in the course of the proof. Introduce the group $\overline{W}$ generated by elements T_i , $i = 1, 2, \dots, n$, subject to the relations $(T_i T_j)^{n_{ij}} = 1$. In particular, one has $n_{ii} = 1$, whence $T_i^2 = 1$. Let $p : \overline{W} \rightarrow W$ be the morphism of groups sending T_i to s_{α_i} . One knows that p is surjective; we shall show that it is injective.

Lemma 5.2. *One can define uniquely, for each $\alpha \in \mathcal{P}(\Delta)$, an element $T_\alpha \in \overline{W}$ in such a way that the following properties hold:*

- (i) $p(T_\alpha) = s_\alpha$,
- (ii) $T_{\alpha_i} = T_i$,
- (iii) *if β and α are two positive roots such that $s_{\alpha_i}(\alpha) = \beta$, then $T_i T_\alpha T_i = T_\beta$.*

⁽²⁰⁾**N.D.E.** The statement has been corrected.

⁽²¹⁾**N.D.E.** Indeed, the hypotheses amount to saying that $(\gamma, \alpha^*) > 0$ and $(\gamma, \beta^*) \geq 0$; then $\gamma - w(\gamma)$ belongs to $\mathbf{N}\alpha + \mathbf{N}\beta$ for every $w \in W_{\alpha, \beta}$, and is $\neq 0$ if $w \notin \{1, s_\beta\}$.

Note first that it follows from 1.2.10 and 3.3.6 that (i) is a consequence of (ii) and (iii), and that (ii) and (iii) determine the T_α completely. We shall make the construction by induction on $\text{ord}_\Delta(\alpha)$. If $\text{ord}_\Delta(\alpha) = 1$, then $\alpha \in \Delta$, and put $T_\alpha = T_i$ if $\alpha = \alpha_i$. Consider the hypothesis:

- (H_p) there exist T_α , for $\alpha \in \mathcal{P}(\Delta)$, $\text{ord}_\Delta(\alpha) \leq p$, satisfying (ii) and condition (iii) whenever $\text{ord}_\Delta(\alpha) \leq p$, $\text{ord}_\Delta(\beta) \leq p$.

This holds for $p = 1$: indeed, if α and $s_{\alpha_i}(\alpha) = \beta$ are simple, α_i and α are orthogonal; thus, putting $\alpha_j = \alpha = \beta$, one has $n_{ij} = 2$, whence

$$T_i T_j T_i = T_j.$$

Suppose $p > 1$ and (H_{p-1}) holds.

A) *Construction of the T_α for $\text{ord}_\Delta(\alpha) \leq p$.* Clearly, it suffices to do this for $\text{ord}_\Delta(\alpha) = p$. There then exists $\alpha_i \in \Delta$ such that $s_{\alpha_i}(\alpha) \in \mathcal{P}(\Delta)$ and $\text{ord}_\Delta(s_{\alpha_i}(\alpha)) < p$ (3.3.3). One will then put

$$(\star) \quad T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

Let us check that T_α depends only on α . Let therefore $\alpha_j \in \Delta$ be such that $s_{\alpha_j}(\alpha) \in \mathcal{P}(\Delta)$ and $\text{ord}_\Delta(s_{\alpha_j}(\alpha)) < p$. Let us prove that

$$(+)\quad T_i T_{s_{\alpha_i}(\alpha)} T_i = T_j T_{s_{\alpha_j}(\alpha)} T_j.$$

Distinguish two cases.

(1) Suppose α a linear combination of α_i and α_j . The same is then true of $s_{\alpha_i}(\alpha)$ and $s_{\alpha_j}(\alpha)$, and, by (H_{p-1}), $T_{s_{\alpha_i}(\alpha)}$ and $T_{s_{\alpha_j}(\alpha)}$ are written as words in T_i and T_j . Since the projection of (+) into W holds, and the theorem is true for $n = 2$, hence p is injective on the subgroup of $\overline{W}$ generated by T_i and T_j , (+) indeed holds.

(2) Suppose α is not a linear combination of α_i and α_j . Then if $w \in W_{\alpha_i, \alpha_j}$, the $w(\alpha)$ are all positive (cf. 3.4.9). The relation to be checked is also written

$$(++)\quad (T_i T_j)^{n_{ij}-1} T_{s_{\alpha_i}(\alpha)} (T_j T_i)^{n_{ij}-1} = T_{s_{\alpha_j}(\alpha)}.$$

Now it follows from 4.4 that the $w(\alpha)$ all have order $< p$ for $w \in W_{\alpha_i, \alpha_j}$, $w \neq 1$. One may therefore apply hypothesis (H_{p-1}) $2(n_{ij} - 1)$ times, and the proof is finished.

B) *Verification of (H_p).*⁽²²⁾ One must check that if $\alpha_j \in \Delta$ and $\beta = s_{\alpha_j}(\alpha)$ satisfies $\text{ord}_\Delta(\beta) \leq p$, then $T_j T_\alpha T_j = T_\beta$. If $\text{ord}_\Delta(\beta) < p$, this follows from what precedes (since $T_j^2 = 1$); thus one may suppose $\text{ord}_\Delta(\beta) = p = \text{ord}_\Delta(\alpha)$. In this case, α and α_j are orthogonal, hence $\beta = \alpha$, and one must see that

$$(\dagger) \quad T_j T_\alpha T_j = T_\alpha.$$

By ($\star$) above, one has $T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i$, and one therefore need only check the following equality:

$$(+++) \quad T_j T_i T_{s_{\alpha_i}(\alpha)} T_i T_j = T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

Put $m = n_{ij}$ and $s_i = s_{\alpha_i}$, $s_j = s_{\alpha_j}$. One has $T_j T_i = (T_i T_j)^{m-1}$ and, by 4.4, $\text{ord}_\Delta(w(\alpha)) < p$ for every $w \in W_{ij}$ distinct from $1 = (s_i s_j)^m$ and from $s_j = s_i (s_i s_j)^{m-1}$. One deduces, by the induction hypothesis, that

$$\begin{aligned} T_j (T_i T_j)^{m-2} T_{s_i(\alpha)} (T_j T_i)^{m-2} T_j &= T_{s_j(s_i s_j)^{m-2} s_i(\alpha)} \\ &= T_{s_i s_j(\alpha)} = T_{s_i(\alpha)} \end{aligned}$$

(the last equality since $s_j(\alpha) = \alpha$), whence finally

$$(T_i T_j)^{m-1} T_{s_i(\alpha)} (T_j T_i)^{m-1} = T_i T_{s_i(\alpha)} T_i,$$

which proves (+++).

⁽²²⁾**N.D.E.** The original has been expanded in what follows.

Lemma 5.3. *Let $h \in \overline{W}$. Write it*

$$h = T_{\alpha_1} \cdots T_{\alpha_m}$$

with $\alpha_i \in \Delta$, not necessarily distinct, in such a way that m is minimal. Then

$$p(h)(\alpha_m) \in -\mathcal{P}(\Delta).$$

Indeed, since $p(T_{\alpha_m})(\alpha_m) = s_{\alpha_m}(\alpha_m) = -\alpha_m$, if $p(h)(\alpha_m)$ were positive, there would exist an index k , $1 \leq k \leq m-1$, such that

$$u = s_{\alpha_{k+1}} \cdots s_{\alpha_m}(\alpha_m) = -s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) \in -\mathcal{P}(\Delta),$$

and $s_{\alpha_k}(u) \in \mathcal{P}(\Delta)$. But then necessarily $u = -\alpha_k$ (3.3.1), whence

$$s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) = \alpha_k,$$

which implies by (iii)

$$T_{\alpha_k} T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}} T_{\alpha_m} = T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}},$$

and this contradicts the minimality of m .

Let now $h \in \overline{W}$ be such that $p(h)(\mathcal{P}(\Delta)) \subset \mathcal{P}(\Delta)$. By lemma 5.3, one has $p(h) = 1$, which proves theorem 5.1 and, in addition:

Corollary 5.4. *If R_+ is a system of positive roots and $w \in W$ is such that $w(R_+) = R_+$, then $w = 1$.*

Corollary 5.5. *The Weyl group acts freely and transitively on the set of systems of positive roots (resp. systems of simple roots, resp. Weyl chambers).*

Now choose a system of simple roots Δ . Put $R^+ = \mathcal{P}(\Delta)$.

⁽²³⁾For every pair of simple roots $(\alpha, \beta) \in \Delta \times \Delta$, denote by $R_{\alpha, \beta}$ the set of roots which are linear combinations of α and β . Put $R_{\alpha, \beta}^+ = R^+ \cap R_{\alpha, \beta}$, and let $W_{\alpha, \beta}$ be the Weyl group of $R_{\alpha, \beta}$, that is, the subgroup of W generated by s_α and s_β .

Theorem 5.6 (Tits). *Let α and β be two simple roots, and $w \in W$ such that $w(\alpha) = \beta$. There exist a sequence of simple roots $\alpha_0, \dots, \alpha_m$ and a sequence $w_0, \dots, w_{m-1}$ of elements of W satisfying the following conditions:*

- (i) $\alpha_0 = \alpha$, $\alpha_m = \beta$.
- (ii) $w = w_{m-1} w_{m-2} \cdots w_0$.
- (iii) $w_i(\alpha_i) = \alpha_{i+1}$ for $0 \leq i \leq m-1$.
- (iv) For every i , $0 \leq i \leq m-1$, such that $\alpha_i \neq \alpha_{i+1}$, one has $w_i \in W_{\alpha_i, \alpha_{i+1}}$.
- (v) For every i , $0 \leq i \leq m-1$, such that $\alpha_i = \alpha_{i+1}$, there exists a simple root β_i such that $w_i \in W_{\alpha_i, \beta_i}$.

Put⁽²⁴⁾

$$M(w) = \text{Card}(R^+ \cap w^{-1}(-R^+)) = \text{Card}\{\alpha \in R^+ \mid w(\alpha) \in -R^+\}.$$

If $M(w) = 0$, then $w(R^+) = R^+$, hence $w = 1$ by 5.4, and the theorem is trivial ($m = 0$, assertions (iii) to (v) are empty). Argue by induction on $M(w)$. If $M(w) > 0$, there exists $\beta_0 \in \Delta$ such that $w(\beta_0) \in -R^+$. Put $\alpha_0 = \alpha$. Consider the set

$$A = w^{-1}(R^+) \cap R_{\alpha_0, \beta_0}.$$

It is a system of positive roots of R_{α_0, β_0} . There therefore exists $w_0 \in W_{\alpha_0, \beta_0}$ such that

$$w_0^{-1}(R_{\alpha_0, \beta_0}^+) = A.$$

⁽²³⁾**N.D.E.** R^+ has been written here instead of R_+ , so as to be able to write $R_{\alpha, \beta}^+ = R_{\alpha, \beta} \cap R^+$.

⁽²⁴⁾**N.D.E.** In what follows, the original has been expanded, and equality (1) corrected.

Put $w' = ww_0^{-1}$. By 3.4.9, one has at once

$$R^+ - R_{\alpha_0, \beta_0}^+ = w_0(R^+ - R_{\alpha_0, \beta_0}^+),$$

whence

$$(1) \quad (R^+ - R_{\alpha_0, \beta_0}^+) \cap w^{-1}(-R^+) = w_0((R^+ - R_{\alpha_0, \beta_0}^+) \cap w'^{-1}(-R^+)).$$

On the other hand,

$$(2) \quad \beta_0 \in R_{\alpha_0, \beta_0}^+ \cap w^{-1}(-R^+),$$

and, since $w_0(R_{\alpha_0, \beta_0}) = R_{\alpha_0, \beta_0}$, one has $w_0(-A) = R_{\alpha_0, \beta_0} \cap w'^{-1}(-R^+)$, whence

$$(2') \quad R_{\alpha_0, \beta_0}^+ \cap w'^{-1}(-R^+) = R_{\alpha_0, \beta_0}^+ \cap w_0(-A) = R_{\alpha_0, \beta_0}^+ \cap -R_{\alpha_0, \beta_0}^+ = \emptyset.$$

It follows from (1), (2) and (2') that $M(w') < M(w)$.

Put $\alpha_1 = w_0(\alpha_0)$; let us show that $\alpha_1 \in \Delta$, i.e. $\alpha_0 \in w_0^{-1}(\Delta)$. One knows that $w(\alpha_0) \in \Delta$, hence $\alpha_0 \in w^{-1}(\Delta)$, hence also $\alpha_0 \in w^{-1}(\Delta) \cap R_{\alpha_0, \beta_0}$, hence α_0 is a simple root of $A = w^{-1}(R^+) \cap R_{\alpha_0, \beta_0} = w_0^{-1}(R_{\alpha_0, \beta_0}^+)$, hence belongs to

$$w_0^{-1}(\Delta \cap R_{\alpha_0, \beta_0}^+) = w_0^{-1}(\{\alpha_0, \beta_0\})$$

(see 3.4.8). Thus $\alpha_1 = w_0(\alpha_0)$ equals α_0 or β_0 . If $\alpha_1 \neq \alpha_0$, one has $\alpha_1 = \beta_0$ and $w_0 \in W_{\alpha_0, \alpha_1}$.

Finally, one has $\beta = w'(\alpha_1)$, with $M(w') < M(w)$, and one concludes by induction.

6. Morphisms of root data

6.1. — Definition. Let $\mathcal{R} = (M, M^*, R, R)$ and $\mathcal{R}' = (M', M'^*, R', R'^*)$ be two root data. Let $f : M' \rightarrow M$ be a linear map and ${}^t f : M^* \rightarrow M'^*$ its transpose.

Definition 6.1.1. One says that f is a *morphism* from $\mathcal{R}'$ to $\mathcal{R}$, and writes $f : \mathcal{R}' \rightarrow \mathcal{R}$, if f induces a bijection from R' onto R and ${}^t f$ a bijection from R^* onto R'^* .

Then ${}^t f$ is a morphism of the dual root data:

$${}^t f : \mathcal{R}^* \rightarrow \mathcal{R}'^*.$$

One easily sees that if f is a morphism from $\mathcal{R}'$ to $\mathcal{R}$, and one puts $\alpha = f(\alpha')$ for $\alpha' \in R'$, then $\alpha'^* = {}^t f(\alpha^*)$. Indeed, one sees immediately that if p and p' denote the maps of 1.2.1 relative to $\mathcal{R}$ and $\mathcal{R}'$, respectively, one has $p' = {}^t f \circ p \circ f$, and the desired assertion follows at once. We leave to the reader the proofs of the following statements, almost all of which are trivial.

Proposition 6.1.2. Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism of root data. If $\alpha' \in R'$ and $\alpha = f(\alpha')$, then $\alpha'^* = {}^t f(\alpha^*)$. Moreover, f induces isomorphisms:

$$R' \xrightarrow{\sim} R, \quad \Gamma_0(R') \xrightarrow{\sim} \Gamma_0(R), \quad \mathcal{V}(R') \xrightarrow{\sim} \mathcal{V}(R),$$

and ${}^t f$ induces isomorphisms:

$$R^* \xrightarrow{\sim} R'^*, \quad \Gamma_0(R^*) \xrightarrow{\sim} \Gamma_0(R'^*), \quad \mathcal{V}(R^*) \xrightarrow{\sim} \mathcal{V}(R'^*),$$

the last being the transpose of the corresponding morphism induced by f . The map $s_{\alpha'} \mapsto s_{f(\alpha')}$ extends to an isomorphism $W(\mathcal{R}') \xrightarrow{\sim} W(\mathcal{R})$ compatible with the actions of these two groups on the sets of 1.1.13.

Proposition 6.1.3. The maps

$$\Delta' \mapsto f(\Delta'), \quad R'_+ \mapsto f(R'_+), \quad C' \mapsto ({}^t f \otimes \mathbf{R})^{-1}(C')$$

define bijective correspondences between systems of simple roots, systems of positive roots, and Weyl chambers for $\mathcal{R}'$ and $\mathcal{R}$. These correspondences are compatible with the action of the Weyl groups and with the correspondences

$$\mathcal{S}(R_+) \leftrightarrow R_+ \leftrightarrow \mathcal{C}(R_+).$$

Lemma 6.1.4. *Morphisms compose. For the morphism $f : \mathcal{R}' \rightarrow \mathcal{R}$ to be an isomorphism, it is necessary and sufficient that $f : M' \rightarrow M$ be bijective.*

6.2. Isogenies

Definition 6.2.1. A morphism $f : \mathcal{R}' \rightarrow \mathcal{R}$ of root data is called an *isogeny* if $f : M' \rightarrow M$ is injective with finite cokernel.

If f is an isogeny, ${}^t f$ is an isogeny.

Definition 6.2.2. Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be an isogeny. Put $K(f) = \text{Coker}(M' \xrightarrow{f} M)$.

Lemma 6.2.3. *One has a natural pairing*

$$K(f) \times K({}^t f) \rightarrow \mathbf{Q}/\mathbf{Z},$$

putting these two finite groups in duality.

This is classical.

Lemma 6.2.4. *If $f : \mathcal{R}' \rightarrow \mathcal{R}$ is a morphism, then $\text{rgss}(\mathcal{R}') = \text{rgss}(\mathcal{R})$. If in addition f is an isogeny, one also has $\text{rgred}(\mathcal{R}') = \text{rgred}(\mathcal{R})$.*

Trivial.

Lemma 6.2.5. *Every morphism of semisimple root data is an isogeny.*

This follows at once from the fact that f must induce an isomorphism from $V' = \mathcal{V}(R')$ onto $V = \mathcal{V}(R)$.

If $\mathcal{R}'$ and $\mathcal{R}$ are semisimple, every isogeny $f : \mathcal{R}' \rightarrow \mathcal{R}$ defines a commutative diagram:

$$\begin{array}{ccc} \Gamma_0(R') & \xrightarrow{\sim} & \Gamma_0(R) \\ \downarrow & & \downarrow \\ M' & \xrightarrow{f} & M. \end{array}$$

If $M = \Gamma_0(R)$, f is necessarily an isomorphism.

Definition 6.2.6. A root datum is called *adjoint* (resp. *simply connected*) if $M = \Gamma_0(R)$, resp. $M^* = \Gamma_0(R^*)$.

An adjoint or simply connected root datum is therefore semisimple. On the other hand, $\mathcal{R}$ is adjoint (resp. simply connected) if and only if $\mathcal{R}^*$ is simply connected (resp. adjoint). By the preceding result, one has:

Proposition 6.2.7. *Let $\mathcal{R}$ be a semisimple root datum. The following conditions are equivalent:*

- (i) $\mathcal{R}$ is adjoint (resp. simply connected).
- (ii) Every isogeny $\mathcal{R}' \rightarrow \mathcal{R}$ (resp. $\mathcal{R} \rightarrow \mathcal{R}'$) is an isomorphism.

Proposition 6.2.8. *Let $\mathcal{R}$ be an adjoint (resp. simply connected) root datum. Every indivisible root (resp. coroot) is an indivisible element of M (resp. M^*).*

Indeed, every indivisible root belongs to a basis of $\Gamma_0(R)$ by 3.3.5.

6.3. Radical and coradical

Let $\mathcal{R}$ be a root datum. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha^* \in R^*\}; \quad N^* = M^*/\mathcal{V}(R^*) \cap M^*.$$

Lemma 6.3.1. *Consider the canonical morphisms:*

$$N \longrightarrow M, \quad M^* \longrightarrow N^*.$$

They are transposes of one another, and N^ is identified with the dual of N .*

This is immediate, taking account of 1.2.5.

Definition 6.3.2. One calls the *coradical* of $\mathcal{R}$, and denotes by $\text{corad}(\mathcal{R})$, the trivial root datum

$$\text{corad}(\mathcal{R}) = (N, N^*, \emptyset, \emptyset).$$

If one puts $\mathcal{R}^0 = (M, M^*, \emptyset, \emptyset)$ (this is a trivial root datum), one therefore has a morphism

$$\text{corad}(\mathcal{R}) \longrightarrow \mathcal{R}^0.$$

Definition 6.3.3. One calls the *radical* of $\mathcal{R}$, and denotes by $\text{rad}(\mathcal{R})$, the trivial root datum:

$$\text{rad}(\mathcal{R}) = \text{corad}(\mathcal{R}^*)^*.$$

(25)

One therefore has a diagram

$$\begin{array}{ccc} \text{corad}(\mathcal{R}) & \xrightarrow{\quad} & \text{rad}(\mathcal{R}) \\ & \searrow \quad \nearrow & \\ & \mathcal{R}^0 & \end{array}$$

whose transpose is the corresponding diagram for $\mathcal{R}^*$.

Lemma 6.3.4. *The canonical morphism $u : \text{corad}(\mathcal{R}) \longrightarrow \text{rad}(\mathcal{R})$ is an isogeny.*

Definition 6.3.5. Put $N(\mathcal{R}) = K(u) = M/((\mathcal{V}(R) \cap M) + \bigcap_{\alpha \in R} \text{Ker}(\alpha^*))$. One then has a canonical pairing

$$N(\mathcal{R}) \times N(\mathcal{R}^*) \longrightarrow \mathbf{Q}/\mathbf{Z}.$$

Lemma 6.3.6. *One has $\text{rgred}(\text{rad}(\mathcal{R})) = \text{rgred}(\text{corad}(\mathcal{R})) = \text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R})$, and the following conditions are equivalent:*

- (i) $\mathcal{R}$ is semisimple,
- (ii) $\text{rad}(\mathcal{R}) = 0$,
- (iii) $\text{corad}(\mathcal{R}) = 0$.

6.4. Products of root data

Definition 6.4.1. Let $\mathcal{R} = (M, M^*, R, R^*)$ and $\mathcal{R}' = (M', M'^*, R', R'^*)$ be two root data. One calls the *product root datum* of $\mathcal{R}$ and $\mathcal{R}'$, and denotes by $\mathcal{R}'' = \mathcal{R} \times \mathcal{R}'$, the root datum (M'', M''^*, R'', R''^*) where

$$\begin{aligned} M'' &= M \times M', & M''^* &= M^* \times M'^*, \\ R'' &= (R \times 0) \cup (0 \times R'), & R''^* &= (R^* \times 0) \cup (0 \times R'^*), \end{aligned}$$

the map $\alpha \mapsto \alpha^*$ being the obvious map.

(25) **N.D.E.** I.e. if one puts $P = \{y \in M^* \mid (y, \alpha) = 0 \text{ for every } \alpha \in R\}$ and $P^* = M/\mathcal{V}(R) \cap M$, one has $\text{rad}(\mathcal{R}) = (P^*, P, \emptyset, \emptyset)$.

Proposition 6.4.2. *Under the preceding conditions, one has canonical isomorphisms*

$$\Gamma_0(R'') \simeq \Gamma_0(R) \times \Gamma_0(R'), \quad \mathcal{V}(R'') \simeq \mathcal{V}(R) \times \mathcal{V}(R'),$$

$$W(\mathcal{R}'') \simeq W(\mathcal{R}) \times W(\mathcal{R}'),$$

etc., and the equalities

$$\text{rgred}(\mathcal{R}'') = \text{rgred}(\mathcal{R}) + \text{rgred}(\mathcal{R}'),$$

$$\text{rgss}(\mathcal{R}'') = \text{rgss}(\mathcal{R}) + \text{rgss}(\mathcal{R}').$$

One also has a canonical isomorphism of root data

$$(\mathcal{R} \times \mathcal{R}')^* \simeq \mathcal{R}^* \times \mathcal{R}'^*.$$

The preceding definitions generalize at once to a product of several factors. One has at once:

Proposition 6.4.3. *Let $\mathcal{R} = \mathcal{R}_1 \times \mathcal{R}_2 \times \cdots \times \mathcal{R}_n$ be a product of root data. The following conditions are equivalent:*

- (i) $\mathcal{R}$ is semisimple (resp. simply connected, resp. adjoint, resp. reduced).
- (ii) Each $\mathcal{R}_i$ is semisimple (resp. simply connected, resp. adjoint, resp. reduced).

Consider the following particular case: let $\mathcal{R}_0$ be a trivial root datum and $\mathcal{R}_1$ a semisimple root datum. One then has a commutative diagram

$$\begin{array}{ccc} & \mathcal{R}_0 \times \mathcal{R}_1 & \\ \nearrow & & \searrow \\ \mathcal{R}_1 & \xrightarrow{\text{id}} & \mathcal{R}_1. \end{array}$$

Lemma 6.4.4. *One has canonical isomorphisms*

$$\begin{array}{ccc} \text{corad}(\mathcal{R}_0 \times \mathcal{R}_1) & \xrightarrow{\sim} & \text{rad}(\mathcal{R}_0 \times \mathcal{R}_1) \\ & \searrow \cong & \nearrow \cong \\ & \mathcal{R}_0 & \end{array}$$

In particular, $N(\mathcal{R}_0 \times \mathcal{R}_1) = 0$.

We shall see later that, conversely, if $N(\mathcal{R}) = 0$, the root datum $\mathcal{R}$ is the product of a semisimple datum and a trivial datum.

6.5. Induced and coinduced root data

Let $\mathcal{R} = (M, M^*, R, R^*)$ be a root datum. Let $N \subset M$ be a subgroup containing the roots, i.e. such that

$$\Gamma_0(R) \subset N \subset M.$$

The canonical linear map $i_N : N \rightarrow M$ gives by transposition a linear map

$${}^t i_N : M^* \rightarrow N^*.$$

Put $R_N = R$ and $R_N^* = {}^t i_N(R^*)$.

Lemma 6.5.1. $\mathcal{R}_N = (N, N^*, R_N, R_N^*)$ is a root datum, and i_N a morphism.

Let us first show that ${}^t i_N$ induces an isomorphism from R^* onto R_N^* . If $\alpha, \beta \in R$ and ${}^t i_N(\alpha^*) = {}^t i_N(\beta^*)$, one has $(\alpha^*, x) = (\beta^*, x)$ for every $x \in N$, in particular for $x \in R$, which gives $\alpha = \beta$ by 1.1.4. The rest follows without difficulty.

Definition 6.5.2. $\mathcal{R}_N$ is called the root datum *induced* by $\mathcal{R}$ on N .

Lemma 6.5.3. *Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism. Put $N = f(M') \subset M$. Then f factors uniquely through i_N .*

In particular, isogenies $\mathcal{R}' \rightarrow \mathcal{R}$, up to isomorphism, correspond bijectively to subgroups of finite index of $M/\Gamma_0(R)$, which makes 6.2.7 more precise.

Let now N^* be a subgroup of M^* containing R^* . One defines the datum *coinduced* by $\mathcal{R}$ on N^* by

$$\mathcal{R}^{N^*} = (\mathcal{R}_{N^*}^*)^*,$$

and one has a canonical morphism:

$$p^{N^*} : \mathcal{R} \rightarrow \mathcal{R}^{N^*}.$$

Lemma 6.5.4. *Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism. There exist subgroups $N \subset M$ and $N'^* \subset M'^*$ such that f factors as*

$$\begin{array}{ccc} \mathcal{R}' & \xrightarrow{f} & \mathcal{R} \\ p^{N'} \downarrow & & \uparrow i_N \\ \mathcal{R}'^{N'^*} & \xrightarrow[\sim]{f_0} & \mathcal{R}_N, \end{array}$$

where f_0 is an isomorphism.

Indeed, take $N = f(M')$ as in 6.5.3. The resulting morphism $M' \rightarrow N$ is surjective, hence its transpose injective. Take the image of the latter as N'^* .

Let us now treat certain particular cases. If one takes $N = \Gamma_0(R)$, one will write $\mathcal{R}_N = \text{ad}(\mathcal{R})$. If one takes $N = \mathcal{V}(R) \cap M$, one will write $\mathcal{R}_N = \text{ss}(\mathcal{R})$. One therefore has a diagram:

$$\text{ad}(\mathcal{R}) \rightarrow \text{ss}(\mathcal{R}) \rightarrow \mathcal{R}.$$

Put $\text{dér}(\mathcal{R}) = \text{ss}(\mathcal{R}^*)^*$ and $\text{sc}(\mathcal{R}) = \text{ad}(\mathcal{R}^*)^*$; by duality, one obtains a diagram:

$$\mathcal{R} \rightarrow \text{dér}(\mathcal{R}) \rightarrow \text{sc}(\mathcal{R}).$$

Proposition 6.5.5.

(i) *In the first row of the diagram*

$$\begin{array}{ccccccc} \text{ad}(\mathcal{R}) & \rightarrow & \text{ss}(\mathcal{R}) & \xrightarrow{\quad} & \text{dér}(\mathcal{R}) & \rightarrow & \text{sc}(\mathcal{R}) \\ & & & \searrow & \nearrow & & \\ & & & \mathcal{R} & & & \end{array}$$

the four data are semisimple and the three morphisms are isogenies.

- (ii) $\text{ad}(\mathcal{R})$ is an adjoint datum, and $\mathcal{R}$ is adjoint if and only if $\text{ad}(\mathcal{R}) \rightarrow \mathcal{R}$ is an isomorphism.
- (iii) $\text{sc}(\mathcal{R})$ is a simply connected datum, and $\mathcal{R}$ is simply connected if and only if $\mathcal{R} \rightarrow \text{sc}(\mathcal{R})$ is an isomorphism.
- (iv) The following conditions are equivalent:
 - (a) $\mathcal{R}$ is semisimple,
 - (b) $\text{ss}(\mathcal{R}) \rightarrow \mathcal{R}$ is an isomorphism,
 - (c) $\mathcal{R} \rightarrow \text{dér}(\mathcal{R})$ is an isomorphism.

Let us pause a moment on the morphism $\text{ss}(\mathcal{R}) \rightarrow \text{dér}(\mathcal{R})$. Returning to the construction of $\text{ss}(\mathcal{R})$ and $\text{dér}(\mathcal{R})$, it is easy to prove:

Lemma 6.5.6. *Let $h : \text{ss}(\mathcal{R}) \rightarrow \text{dér}(\mathcal{R})$ be the canonical isogeny. One has $K(h) \simeq N(\mathcal{R})$.*

6.5.7. — Consider other particular cases of induced data. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for } \alpha \in R\} \times \Gamma_0(R);$$

one knows that the sum is direct by 1.2.5. It follows that the root datum $\mathcal{R}_N$ is identified with the product $\text{ad}(\mathcal{R}) \times \text{corad}(\mathcal{R})$.

One can do the same replacing $\Gamma_0(R)$ by $\mathcal{V}(R) \cap M$, then dualize these two constructions. One thus obtains a diagram of root data:

$$\begin{array}{ccccccc}
 \text{ad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) & \longrightarrow & \text{dér}(\mathcal{R}) & \longrightarrow & \text{sc}(\mathcal{R}) & ; \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & \\
 \text{ad}(\mathcal{R}) \times \text{corad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) & \longrightarrow & \mathcal{R} & \longrightarrow & \text{dér}(\mathcal{R}) \times \text{rad}(\mathcal{R}) & \longrightarrow \text{sc}(\mathcal{R}) \times \text{rad}(\mathcal{R}) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & \\
 \text{ad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) & \longrightarrow & \text{dér}(\mathcal{R}) & \longrightarrow & \text{sc}(\mathcal{R}) &
 \end{array}$$

which is commutative, as one checks at once. This diagram is self-dual in an obvious sense. The horizontal morphisms are isogenies. The composites of the vertical arrows are the identity.

Lemma 6.5.8. *Let h_1 and h_2 be the canonical isogenies:*

$$\text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) \xrightarrow{h_1} \mathcal{R} \xrightarrow{h_2} \text{dér}(\mathcal{R}) \times \text{rad}(\mathcal{R}).$$

One has $K(h_1) \simeq K(h_2) \simeq N(\mathcal{R})$.

This is trivial from the definitions.

Corollary 6.5.9. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

- (i) $N(\mathcal{R}) = 0$, i.e. $\text{corad}(\mathcal{R}) \longrightarrow \text{rad}(\mathcal{R})$ is an isomorphism.
- (ii) $h : \text{ss}(\mathcal{R}) \longrightarrow \text{dér}(\mathcal{R})$ is an isomorphism.
- (iii) $h_1 : \text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) \longrightarrow \mathcal{R}$ is an isomorphism.
- (iv) $h_2 : \mathcal{R} \longrightarrow \text{dér}(\mathcal{R}) \times \text{rad}(\mathcal{R})$ is an isomorphism.
- (v) $\mathcal{R}$ is the product of a semisimple datum and a trivial datum.

Let us also state a trivial consequence of the preceding remarks:

Corollary 6.5.10. *For every root datum $\mathcal{R}$, there exist isogenies*

$$\text{ad}(\mathcal{R}) \times \mathcal{R}_0 \longrightarrow \mathcal{R} \longrightarrow \text{sc}(\mathcal{R}) \times \mathcal{R}_0,$$

where $\mathcal{R}_0$ is “the” trivial root datum of rank $\text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R})$.

Finally, let us point out a result which may be useful:

Lemma 6.5.11. *Let $\mathcal{R}$ be a root datum, Δ a system of simple roots, Δ' a subset of Δ ; consider the root datum (cf. 3.4.7)*

$$\mathcal{R}_{\Delta'} = (M, M^*, R_{\Delta'}, R_{\Delta'}^*).$$

- (i) *If $\mathcal{R}$ is simply connected, $\text{dér}(\mathcal{R}_{\Delta'})$ is simply connected.*
- (ii) *If $\mathcal{R}$ is adjoint, $\text{ss}(\mathcal{R}_{\Delta'})$ is adjoint.*

The two assertions are clearly equivalent by duality. The second reduces to checking the formula:

$$M \cap \mathcal{V}(R_{\Delta'}) = \Gamma_0(R_{\Delta'});$$

now, if $M = \Gamma_0(R)$, both sides are equal to the subgroup of M generated by Δ' .

6.6. Weights

Definition 6.6.1. Let $\mathcal{R}$ be a root datum. Put⁽²⁶⁾

$$\Lambda(\mathcal{R}) = \{x \in V(R) \mid (\alpha^*, x) \in \mathbf{Z} \text{ for every } \alpha^* \in R^*\}.$$

The elements of $\Lambda(\mathcal{R})$ are called the *weights* of $\mathcal{R}$. The weights of $\mathcal{R}^*$ are called the *coweights* of $\mathcal{R}$.

One has $\Gamma_0(R) \subset \Lambda(\mathcal{R})$, and $\Lambda(\mathcal{R})$ is stable under $W(\mathcal{R})$.

Lemma 6.6.2. The bilinear map $V^* \times V \longrightarrow \mathbf{Q}$ induces a duality

$$\Gamma_0(R^*) \times \Lambda(\mathcal{R}) \longrightarrow \mathbf{Z}.$$

Trivial.

Corollary 6.6.3. Let $\Delta^* = (\alpha_1^*, \dots, \alpha_n^*)$ be a system of simple coroots. Let $p_i, i = 1, 2, \dots, n$, be the elements of $\mathcal{V}(R)$ defined by

$$(\alpha_i^*, p_j) = \delta_{ij},$$

(whence $s_{\alpha_i}(p_i) = p_i - \alpha_i$ and $s_{\alpha_i}(p_j) = p_j$ for $i \neq j$).^(*) Then $\Lambda(\mathcal{R})$ is the free abelian group generated by the p_i .

The p_i are called the *fundamental weights* corresponding to the system of simple coroots Δ^* .

Corollary 6.6.4. For every $\alpha^* \in \Delta^*$, one therefore has $(\alpha^*, \sum_i p_i) = 1$, hence $\sum_i p_i = \rho_{R_+}$ (cf. 3.5.1), where $R_+ = \mathcal{P}(\text{ind}(\Delta))$.

Corollary 6.6.5. For every $x \in \mathcal{V}(R)$, one has $x = \sum_i (\alpha_i^*, x) p_i$.

Note that $R^* \subset \Gamma_0(R^*)$ and $R \subset \Lambda(\mathcal{R})$, hence $(\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*)$ is a root datum.

Corollary 6.6.6. The canonical morphism $\Gamma_0(R^*) \longrightarrow M^*$ is the transpose of the morphism $x \longmapsto \sum_i (\alpha_i^*, x) p_i$, which defines a morphism of root data, and one has a commutative diagram:

$$\begin{array}{ccc} & & \text{sc}(\mathcal{R}) \\ & \nearrow & \downarrow \simeq \\ \mathcal{R} & \longrightarrow & (\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*). \end{array}$$

One therefore has an explicit description of $\text{sc}(\mathcal{R})$ in terms of the weights of $\mathcal{R}$. Similarly, one finds a commutative diagram:

$$\begin{array}{ccc} \text{ad}(\mathcal{R}) & & \\ \downarrow \simeq & \searrow & \\ (\Gamma_0(R), \Lambda(\mathcal{R}^*), R, R^*) & \longrightarrow & \mathcal{R} \end{array}$$

Corollary 6.6.7. For $\mathcal{R}$ to be simply connected, it is necessary and sufficient that $M = \Lambda(\mathcal{R})$.

Remark 6.6.8. One has $\Lambda(\mathcal{R}) \cap M = \mathcal{V}(R) \cap M$. For $\mathcal{R}$ to be semisimple, it is therefore necessary and sufficient that $M \subset \Lambda(\mathcal{R})$.

It also follows from the results of 6.5 that:

Corollary 6.6.9. For $\mathcal{R}$ to be the product of a simply connected datum and a trivial datum, it is necessary and sufficient that $M \supset \Lambda(\mathcal{R})$.

⁽²⁶⁾**N.D.E.** $\Gamma(\mathcal{R})$ has been replaced by $\Lambda(\mathcal{R})$ to avoid any risk of confusion with $\Gamma_0(R)$.

^(*)Attention: if the datum is not reduced, the α_i do not form a system of simple roots.

Consider now the canonical isogeny

$$f : \text{ad}(\mathcal{R}) \longrightarrow \text{sc}(\mathcal{R}),$$

and put $Z(\mathcal{R}) = K(f)$. One has $Z(\mathcal{R}) \simeq Z(\text{sc } \mathcal{R}) \simeq Z(\text{ad } \mathcal{R})$.

Corollary 6.6.10. *One has a canonical isomorphism $Z(\mathcal{R}) = \Lambda(\mathcal{R})/\Gamma_0(R)$. More precisely, one has an exact sequence of $W(\mathcal{R})$ -modules:*

$$0 \longrightarrow \Gamma_0(R) \longrightarrow \Lambda(\mathcal{R}) \longrightarrow Z(\mathcal{R}) \longrightarrow 0.$$

Corollary 6.6.11. *One has a canonical pairing*

$$Z(\mathcal{R}^*) \times Z(\mathcal{R}) \longrightarrow \mathbf{Q}/\mathbf{Z}$$

putting these groups in duality.

Remark 6.6.12. One has $Z(\mathcal{R} \times \mathcal{R}') \simeq Z(\mathcal{R}) \times Z(\mathcal{R}')$. Consider, in particular, simply connected data $\mathcal{R}_i$, $i = 1, 2, \dots, n$, and a trivial datum $\mathcal{R}_0$. Put $\mathcal{R} = \mathcal{R}_0 \times \mathcal{R}_1 \times \dots \times \mathcal{R}_n$. Let $\mathcal{R} = (M, M^*, R, R^*)$, $\mathcal{R}_0 = (M_0^*, M_0^*, \emptyset, \emptyset)$. One has

$$M/\Gamma_0(R) \simeq M_0 \times Z(\mathcal{R}_1) \times \dots \times Z(\mathcal{R}_n).$$

6.7. Automorphisms

An automorphism of $\mathcal{R}$ is, by 6.1.4, an automorphism u of M such that $u(R) = R$, ${}^t u(R^*) = R^*$. In particular, every element w of $W(\mathcal{R})$ defines an automorphism of $\mathcal{R}$.

Lemma 6.7.1. *$W(\mathcal{R})$ is an invariant subgroup of $\text{Aut}(\mathcal{R})$. More precisely, if $u \in \text{Aut}(\mathcal{R})$ and $\alpha \in R$, one has*

$$us_\alpha u^{-1} = s_{u(\alpha)}.$$

The proof is the same as that of 1.2.10.

Proposition 6.7.2. *Let Δ be a system of simple roots. Put*

$$E_\Delta(\mathcal{R}) = \{u \in \text{Aut}(\mathcal{R}) \mid u(\Delta) = \Delta\}.$$

Then $\text{Aut}(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta(\mathcal{R})$.

This follows at once from the fact that $W(\mathcal{R})$ acts simply transitively on the systems of simple roots, and that if Δ is a system of simple roots of R , $u(\Delta)$ is a system of simple roots for every automorphism u of $\mathcal{R}$.

We shall see later a simpler description of $E_\Delta(\mathcal{R})$ in the case of reduced irreducible root data.⁽²⁷⁾

Definition 6.7.3. Denote by $\text{Aut}^s(\mathcal{R})$ ⁽²⁸⁾ the set of $u \in \text{Aut}(\mathcal{R})$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{u} & \mathcal{R} \\ & \searrow & \swarrow \\ & \text{rad}(\mathcal{R}) & \end{array}$$

Put $E_\Delta^s(\mathcal{R}) = E_\Delta(\mathcal{R}) \cap \text{Aut}^s(\mathcal{R})$.

Remark 6.7.4. If $u \in \text{Aut}(\mathcal{R})$, one therefore has $u \in \text{Aut}^s(\mathcal{R})$ if and only if $(u - \text{id})(M) \subset \mathcal{V}(R)$. In particular, $W(\mathcal{R}) \subset \text{Aut}^s(\mathcal{R})$. It follows at once that:

Proposition 6.7.5. *The group $\text{Aut}^s(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta^s(\mathcal{R})$, for every system of simple roots Δ .*

⁽²⁷⁾**N.D.E.** When $\mathcal{R}$ is simply connected or adjoint, see 7.4.5.

⁽²⁸⁾**N.D.E.** The exponent s is intended to suggest “semisimple”.

With every automorphism of $\mathcal{R}$ there is associated by functoriality an automorphism of $\text{ad}(\mathcal{R})$. One therefore has a canonical morphism

$$\text{Aut}(\mathcal{R}) \longrightarrow \text{Aut}(\text{ad}(\mathcal{R})).$$

Lemma 6.7.6. *The morphism $\text{Aut}^s(\mathcal{R}) \longrightarrow \text{Aut}(\text{ad}(\mathcal{R}))$ is injective.*

Indeed, let u be an automorphism of M such that $(u - \text{id})(M) \subset \mathcal{V}(R)$ and ${}^t u(\alpha^*) = \alpha^*$ for $\alpha^* \in R^*$. For every $x \in M$, one has

$$(\alpha^*, u(x) - x) = ({}^t u(\alpha^*) - \alpha^*, x) = 0,$$

hence $u(x) - x = 0$ by 1.2.5.

Lemma 6.7.7. *The group $\text{Aut}^s(\mathcal{R})$ is finite.*

Indeed, it suffices to prove that $\text{Aut}(\mathcal{R})$ is finite if $\mathcal{R}$ is adjoint. Since M is then generated by R , every automorphism of $\mathcal{R}$ is determined by the permutation of R which it defines.

Remark 6.7.8. One sees at once that $\text{Aut}(\mathcal{R})$ (resp. $E_\Delta(\mathcal{R})$) is finite if and only if $\text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R}) \leq 1$.

6.8. p -morphisms of reduced root data

In this section, p is an integer > 0 , fixed once and for all.

Definition 6.8.1. Let $\mathcal{R} = (M, M^*, R, R^*)$ and $\mathcal{R}' = (M', M'^*, R', R'^*)$ be two reduced root data. One says that a morphism of groups

$$f : M' \longrightarrow M$$

is a p -morphism from $\mathcal{R}'$ to $\mathcal{R}$ if the following conditions hold: there exist a bijection

$$u : R \xrightarrow{\sim} R'$$

and a map $q : R \longrightarrow \{p^n, n \in \mathbf{N}\}$ such that:

- (i) $f(u(\alpha)) = q(\alpha)\alpha$ for every $\alpha \in R$.
- (ii) ${}^t f(\alpha^*) = q(\alpha)u(\alpha)^*$ for every $\alpha \in R$.⁽²⁹⁾

Corollary 6.8.2. *A 1-morphism is precisely a morphism.*

Corollary 6.8.3. *The transpose of a p -morphism is a p -morphism.*

Lemma 6.8.4. *If $w \in W(\mathcal{R})$, $\alpha \in R$, one has $q(w(\alpha)) = q(\alpha)$. The map $s_\alpha \longmapsto s_{u(\alpha)}$ extends to an isomorphism $\bar{u} : W(\mathcal{R}) \longrightarrow W(\mathcal{R}')$ such that*

$$u(w(\alpha)) = \bar{u}(w)(u(\alpha)).$$

It suffices to prove that for $\alpha, \beta \in R$, one has $u(s_\alpha(\beta)) = s_{u(\alpha)}u(\beta)$ and $q(s_\alpha(\beta)) = q(\beta)$. Now one has successively:

$$\begin{aligned} f(s_{u(\alpha)}u(\beta)) &= f(u(\beta)) - (u(\alpha)^*, u(\beta))f(u(\alpha)) \\ &= q(\beta)\beta - q(\beta)q(\alpha)^{-1}(\alpha^*, \beta)q(\alpha)\alpha \\ &= q(\beta)(\beta - (\alpha^*, \beta)\alpha) = q(\beta)s_\alpha(\beta). \end{aligned}$$

If $\gamma = u^{-1}(s_{u(\alpha)}u(\beta))$, one therefore has $q(\gamma)\gamma = f(u(\gamma)) = q(\beta)s_\alpha(\beta)$. The two roots γ and $s_\alpha(\beta)$ are therefore proportional (over $\mathbf{Q}$), hence equal or opposite, but $q(\gamma)$ and $q(\beta)$ are positive. One therefore has $q(\gamma) = q(\beta)$ and $\gamma = s_\alpha(\beta)$.

Definition 6.8.5. The $q(\alpha)$ are called the *root exponents* of f .

⁽²⁹⁾**N.D.E.** Note that these two conditions imply $q(\alpha)(u(\alpha^*), u(\beta)) = q(\beta)(\alpha^*, \beta)$ for all $\alpha, \beta \in R$.

Example 6.8.6. Let $\mathcal{R}$ be a reduced root datum and $q = p^n$ ($n \in \mathbf{N}$). Then multiplication by $q : M \rightarrow M$, $x \mapsto qx$, is a p -morphism all of whose root exponents are q (and $u = \text{id}$); it is denoted

$$q : \mathcal{R} \rightarrow \mathcal{R}.$$

Proposition 6.8.7. *With the notations of 6.8.1, u gives an isomorphism from the set of systems of simple roots (resp. positive roots) of R onto the corresponding set for R' .*

This follows at once from 3.1.5 (resp. 3.2.1).

7. Structure

7.1. Decomposition of a root datum

Proposition 7.1.1. *Let $\mathcal{R}$ be a root datum, Δ a system of simple roots.*

- (i) *Let R' and R'' be two closed symmetric sets of roots forming a partition of R . If one puts $\Delta' = \Delta \cap R'$, $\Delta'' = \Delta \cap R''$, then $R' = R_{\Delta'}$, $R'' = R_{\Delta''}$, and every root of Δ' is orthogonal to every root of Δ'' .*
- (ii) *Let Δ' and Δ'' be two orthogonal subsets of Δ forming a partition of Δ . Then $R' = R_{\Delta'}$ and $R'' = R_{\Delta''}$ form a partition of R .*

Let us first prove (i).

Lemma 7.1.2. *Under the conditions of (i), if α , β and $\alpha + \beta$ are roots, all three belong to R' or all three to R'' .*

Suppose, for example, $\alpha + \beta \in R'$. Then one cannot have $\alpha, \beta \in R''$, since R'' is closed; suppose therefore $\alpha \in R'$. Then $-\alpha \in R'$, and $\beta = (\beta + \alpha) - \alpha \in R'$.

Let us now show that $R' = R_{\Delta'}$ by induction on the order of a positive root $\alpha \in R' \cap \mathcal{P}(\Delta)$. If $\text{ord}_{\Delta}(\alpha) = 1$, then $\alpha \in R' \cap \Delta = \Delta'$. If $\text{ord}_{\Delta}(\alpha) > 1$, there exists $\beta \in \Delta$ such that $\alpha - \beta \in R$. By the lemma, one has $\beta \in \Delta'$, $\alpha - \beta \in R'$, hence $\alpha - \beta \in R_{\Delta'}$ by induction, and finally $\alpha = (\alpha - \beta) + \beta \in R_{\Delta'}$.

Finally, let us show that Δ' and Δ'' are orthogonal. If $\alpha \in \Delta'$ and $\beta \in \Delta''$, then $(\beta^*, \alpha) \leq 0$. If $(\beta^*, \alpha) \neq 0$, $\beta + \alpha$ is a root, contrary to the lemma.

Let us prove (ii). If Δ' or Δ'' is empty, this is immediate. Otherwise, if $R_{\Delta'}$ and $R_{\Delta''}$ do not form a partition of R , there exists a root α of the form

$$\alpha = \sum m'_i \alpha'_i + \sum m''_j \alpha''_j, \quad m'_i \in \mathbf{Z}_+, \quad m''_j \in \mathbf{Z}_+,$$

where α'_i (resp. α''_j) denote elements of Δ' (resp. Δ''). Applying 3.1.2, one deduces a relation of the form (interchanging Δ' and Δ'' if necessary):

$$\delta = \gamma + \beta, \quad \gamma \in R_{\Delta'}, \quad \beta \in \Delta'', \quad \delta \in R.$$

But since $(\beta^*, \gamma) = 0$, $\gamma - \beta$ is also a root by 2.2.5, which is impossible.

Proposition 7.1.3. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

- (i) *There exists no nontrivial partition of R into two closed symmetric subsets.*
- (ii) *For one (resp. every) system of simple roots Δ of R , there exists no partition of Δ into two nonempty orthogonal subsets.*
- (iii) *The natural representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible.*
- (iv) *For every pair (α, β) of roots, there exists a sequence of roots $\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_n$, with $\alpha = \alpha_0$, $\alpha_n = \beta$, the roots α_i and α_{i+1} ($i = 0, \dots, n-1$) being nonorthogonal.*

One has (i) $\Leftrightarrow$ (ii) by 7.1.1. Clearly (iv) $\Rightarrow$ (ii). Conversely, if (ii) holds for Δ , condition (iv) holds if $\alpha, \beta \in \Delta$. Now, for every root there exists a simple root not orthogonal to it

(3.1.1, for example). On the other hand, (iii) $\Rightarrow$ (i). Indeed, under the conditions of 7.1.1, $\mathcal{V}(R')$ is stable under $W(\mathcal{R})$. It remains to prove (i) $\Rightarrow$ (iii).

Let therefore H be a vector subspace of $\mathcal{V}(R)$ stable under $W(\mathcal{R})$. For every $\alpha \in R$, the equation $s_\alpha(H) = H$ gives at once $\alpha \in H$ or $\alpha^* \in H^\perp$ (the orthogonal of H in $\mathcal{V}(R^*)$, which is in duality with $\mathcal{V}(R)$). Putting $R' = \{\alpha \in R \mid \alpha \in H\}$ and $R'' = \{\alpha \in R \mid \alpha^* \in H^\perp\}$, one has obtained a partition of R into two closed symmetric subsets.

Definition 7.1.4. A root datum (resp. root system) satisfying the equivalent conditions of 7.1.3 and of semisimple rank $\neq 0$ is called *irreducible*.

Corollary 7.1.5. For every root datum $\mathcal{R}$, there exists a unique partition (up to order) of R into closed symmetric irreducible subsets.

Corollary 7.1.6. Every adjoint (resp. simply connected) root datum is a product of irreducible adjoint (resp. simply connected) root data.

It suffices to see this in the adjoint case. The assertion then follows from the fact that under the conditions of 7.1.1 one has

$$\Gamma_0(R) = \Gamma_0(R') \times \Gamma_0(R'').$$

Corollary 7.1.7. For every root datum (resp. reduced root datum) $\mathcal{R}$, there exists an isogeny $\mathcal{R} \rightarrow \mathcal{R}'$, where $\mathcal{R}'$ is a product of a trivial root datum and irreducible (resp. reduced) simply connected root data.

7.2. Properties of irreducible root data

Definition 7.2.1. Let $\mathcal{R}$ be an irreducible root datum. For every $\alpha \in R$, put

$$\text{long}(\alpha) = \ell(\alpha)/\ell(\alpha_0);$$

where $\alpha_0 \in R$ is such that $\ell(\alpha_0)$ is minimal; one says that $\text{long}(\alpha)$ is the *length* of α .

Lemma 7.2.2. Let $\mathcal{R}$ be an irreducible root datum. The Weyl group acts transitively on the set of roots of the same length.

Indeed, let $\alpha, \beta \in R$. Since the representation of W on $\mathcal{V}(R)$ is irreducible, α cannot be orthogonal to all the $w(\beta)$, $w \in W$. There therefore exists $w \in W$ with $w(\beta)$ nonorthogonal to α . Now $\ell(w(\beta)) = \ell(\beta)$, and one concludes by 2.3.2.

Lemma 7.2.3. If $\mathcal{R}$ is irreducible and reduced, $\text{long}(R)$ is $\{1\}$, $\{1, 2\}$ or $\{1, 3\}$.

By the remark used above, for every $\alpha, \beta \in R$ there always exists $w \in W$ such that $w(\beta)$ is not orthogonal to α . One therefore has $\ell(\alpha)/\ell(\beta) = 1, 2, 3, 1/2$ or $1/3$ (by 2.3.1). One therefore has $\text{long}(\alpha) = 1, 2$ or 3 , but if $\text{long}(\alpha) = 2$, $\text{long}(\beta) = 3$, then $\ell(\alpha)/\ell(\beta) = 2/3$, which is impossible.

Remark 7.2.4. Reasoning similarly, one proves the following result: if R is irreducible and nonreduced with $\text{rgss}(\mathcal{R}) > 1$, one has $\text{long}(R) = \{1, 2, 4\}$. Putting $\text{long}^{-1}(i) = R_i$, one has $\text{ind}(R) = R_1 \cup R_2$, $R_4 = 2R_1$, and two nonproportional roots of R_1 are orthogonal. Conversely, if R is an irreducible reduced system such that $\text{long}(R) = \{1, 2\}$, put $\text{long}^{-1}(i) = R_i$ and suppose that two nonproportional roots of R_1 are orthogonal; then $R \cup 2R_1$ is irreducible, nonreduced, and $\text{ind}(R \cup 2R_1) = R$.

Lemma 7.2.5. If $\mathcal{R}$ is an irreducible root datum, $\mathcal{R}^*$ is also one, and the product

$$\text{long}(\alpha) \text{long}(\alpha^*)$$

is constant as α runs through R .

This follows at once from 7.1.3 (iv) and 2.2.6.

Definition 7.2.6. Let $\mathcal{R}$ be any root datum. One calls the *length* of $\alpha \in R$, and denotes by $\text{long}(\alpha)$, the length of α in its irreducible component.

Lemma 7.2.7. *There exists a unique group homomorphism $u : \Gamma_0(R) \longrightarrow \Gamma_0(R^*)$ such that $u(\alpha) = \text{long}(\alpha)\alpha^*$ for $\alpha \in R$.*

By 3.5.5, it suffices to check that if $\alpha, \beta, \alpha + \beta \in R$, one has

$$\text{long}(\alpha)\alpha^* + \text{long}(\beta)\beta^* = \text{long}(\alpha + \beta)(\alpha + \beta)^*.$$

But α, β and $\alpha + \beta$ are in the same irreducible component of R by 7.1.2, and one is reduced to 1.2.2.

Remark 7.2.8. Let u be as in 7.2.7. For $\alpha, \beta \in R$, one has $(u(\alpha), \beta) = (u(\beta), \alpha)$. Indeed, this amounts to seeing that

$$\text{long}(\alpha)(\alpha^*, \beta) = \text{long}(\beta)(\beta^*, \alpha),$$

which clearly holds if α and β are orthogonal. If α and β are not orthogonal, they are in the same irreducible component of R , and one is reduced to 1.2.1, formula (9).

Remark 7.2.9. The symmetric bilinear form $(u(x), y)$ is positive nondegenerate on $\Gamma_0(R)$.

Indeed, let R_i be the irreducible components of R . One has

$$\Gamma_0(R) = \prod_i \Gamma_0(R_i),$$

and the bilinear form $(u(x), y)$ is the product of the forms

$$2^{-1}\ell(\alpha_i)^{-1}(p(x), y)$$

on the $\Gamma_0(R_i)$, where $\ell(\alpha_i)$ is the minimum of $\ell(\alpha)$ for $\alpha \in R_i$. Now these various symmetric bilinear forms are positive nondegenerate (1.2.6).

7.3. Cartan matrix

Let $\mathcal{R}$ be a root datum. If Δ is a system of simple roots, one calls the *Cartan matrix* of $\mathcal{R}$ relative to Δ the square matrix indexed by Δ defined by

$$a_{\alpha, \beta} = (\alpha^*, \beta) \quad \text{for } \alpha, \beta \in \Delta.$$

Note first that if Δ' is another system of simple roots and w an element of $W(\mathcal{R})$ such that $w(\Delta) = \Delta'$, one has

$$(w(\alpha)^*, w(\beta)) = (\alpha^*, \beta),$$

hence the Cartan matrix of $\mathcal{R}$ relative to Δ' is obtained from that relative to Δ by the isomorphism $\Delta \longrightarrow \Delta'$ of index sets defined by w . It follows that, up to an isomorphism of the index set, the Cartan matrix depends only on $\mathcal{R}$.

Proposition 7.3.1. *The Cartan matrix has the following properties:*

- (i) $a_{\alpha, \alpha} = 2$, $a_{\alpha, \beta} \leq 0$ for $\alpha \neq \beta$.
- (ii) $a_{\alpha, \beta} = 0$ implies $a_{\beta, \alpha} = 0$.
- (iii) *There exist strictly positive integers m_α ($= \text{long}(\alpha)$) such that the matrix $(m_\alpha a_{\alpha, \beta})$ is symmetric, positive and nondegenerate.*
- (iv) *The diagonal minors of the matrix $(a_{\alpha, \beta})_{\alpha, \beta \in \Delta}$, i.e. the determinants*

$$\det(a_{\alpha, \beta})_{\alpha, \beta \in \Delta'} \quad \text{for } \Delta' \subset \Delta,$$

are strictly positive.

- (v) *One has $s_\alpha(\beta) = \beta - a_{\alpha, \beta}\alpha$ and $s_\alpha(\beta^*) = \beta^* - a_{\beta, \alpha}\alpha^*$.*

Indeed, (v) is a definition, (i) follows from 3.2.11, (ii) from 2.2.2, (iii) from 7.2.9, and (iv) follows at once from (iii) by the relation

$$\det(m_\alpha a_{\alpha,\beta})_{\alpha,\beta \in \Delta'} = \left(\prod_{\alpha \in \Delta'} m_\alpha \right) \det(a_{\alpha,\beta})_{\alpha,\beta \in \Delta'}.$$

Proposition 7.3.2. *Let $\mathcal{R}$ and $\mathcal{R}'$ be two simply connected (resp. adjoint) reduced root data, Δ (resp. Δ') a system of simple roots of $\mathcal{R}$ (resp. $\mathcal{R}'$), and $u : \Delta \rightarrow \Delta'$ an isomorphism such that if $(a_{\alpha,\beta})$ and $(a'_{\alpha',\beta'})$ denote the Cartan matrices of $\mathcal{R}$ and $\mathcal{R}'$ relative to Δ and Δ' , one has:*

$$a'_{u(\alpha),u(\beta)} = a_{\alpha,\beta}.$$

Then there exists a unique isomorphism from $\mathcal{R}$ onto $\mathcal{R}'$ inducing u on Δ .

Clearly, it suffices to give the proof in the adjoint case. Then $M = \Gamma_0(R)$ and $M' = \Gamma_0(R')$ are the free abelian groups generated by Δ and Δ' . There therefore exists a unique group isomorphism from M onto M' inducing u on Δ . Denote it also by u . Let us show that $u(R) \subset R'$. Every root α of $\mathcal{R}$ is written $s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1})$, with $\alpha_i \in \Delta$. Clearly,

$$u(\alpha) = s_{u(\alpha_1)} \cdots s_{u(\alpha_n)}(u(\alpha_{n+1})),$$

by the hypothesis on u and relations (v) of 7.3.1. It remains to prove that ${}^t u(R'^*) \subset R^*$, which follows from the fact that the elements of M^* (resp. M'^*) are determined by duality with R or Δ (resp. R' or Δ'), by 1.2.5.

Corollary 7.3.3. *A reduced simply connected or adjoint root datum is determined up to isomorphism by its Cartan matrix.*

Corollary 7.3.4. *Let $\mathcal{R}$ be a reduced simply connected (resp. adjoint) root datum, and Δ a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the set Δ which leave the Cartan matrix invariant.*

Remark 7.3.5. The question of the existence of a root datum corresponding to a given Cartan matrix satisfying (i), (ii) and (iv) (for example) is not easily solved directly, without using the classification.

7.4. Dynkin diagrams

Definition 7.4.1. A *Dynkin diagram structure* (the word “scheme” has been banished for obvious reasons) on a finite set Δ is the datum of a set of pairs of distinct elements of Δ , called *linked pairs*, and of a map from Δ to the set $\{1, 2, 3\}$. The notion of isomorphism of such structures is obvious.

Definition 7.4.2. Let $\mathcal{R}$ be a root datum and Δ a system of simple roots. One calls the *Dynkin diagram* of $\mathcal{R}$ relative to Δ the set Δ , two simple roots being linked if and only if they are not orthogonal, each root being assigned its length.

Proposition 7.4.3. *Dynkin diagram and Cartan matrix determine one another bijectively.*

Indeed, the equivalence

$$\alpha \text{ and } \beta \text{ are not linked} \iff a_{\alpha,\beta} = 0,$$

and the relation

$$\text{long}(\alpha)a_{\alpha,\beta} = \text{long}(\beta)a_{\beta,\alpha}$$

(with $\inf \text{long}(\alpha) = 1$ in each connected component of the diagram) determine the $a_{\alpha,\beta}$ in terms of the links and lengths, and conversely (the details of the verification are left to the reader).

Corollary 7.4.5. *Let $\mathcal{R}$ be a reduced simply connected or adjoint root datum and Δ a system of simple roots. The group $E_{\Delta}(\mathcal{R})$ is identified with the automorphism group of the Dynkin diagram of $\mathcal{R}$ relative to Δ , that is, with the group of permutations of Δ preserving lengths and links.*

$$\overset{1}{\bullet} - \overset{3}{\bullet} \quad G_2.$$
$$E_{\wedge}(\mathcal{R}) = \mathfrak{S}_3 \quad \text{for } D_4.$$

Let $f : \mathcal{R} \rightarrow \mathcal{R}'$ be a p -morphism (cf. 6.8). It is clear from the definitions that the associated bijection $u : R \xrightarrow{\sim} R'$ makes the systems of simple roots, systems of positive roots, irreducible components (etc.) of R and R' correspond. Suppose therefore, for simplicity, that R and R' are *irreducible*.

$$\text{long}(u(\alpha))(u(\alpha)^*, u(\beta)) = \text{long}(u(\beta))(u(\beta)^*, u(\alpha)).$$

⁽³⁰⁾N.D.E. For a slightly different proof, see also [De80].

One deduces at once that for nonorthogonal α and β , one has

$$\frac{q(\alpha)^2 \text{long}(\alpha)}{\text{long}(u(\alpha))} = \frac{q(\beta)^2 \text{long}(\beta)}{\text{long}(u(\beta))},$$

and one then concludes by 7.1.3 (iv).

Remark 7.5.2. It follows from 7.2.2 and 6.8.4 that $q(\alpha)$ depends only on $\text{long}(\alpha)$. One then easily sees that if $q(\alpha)$ is not constant, $q(\alpha) \text{long}(\alpha)$ is constant, which shows that then $p = 2$ or 3 . A glance at the diagrams of the preceding section shows that there are four possible cases (the same letter denotes a Dynkin diagram and the corresponding reduced simply connected root datum):

$$\begin{aligned} p = 2, \quad B_n &\xrightarrow{f_1} C_n, \quad C_n \xrightarrow{f_2} B_n \quad (\text{with } C_2 = B_2). \\ p = 2, \quad F_4 &\xrightarrow{g} F_4. \\ p = 3, \quad G_2 &\xrightarrow{h} G_2. \end{aligned}$$

The reader will note that $f_1 \circ f_2$, $f_2 \circ f_1$, $g \circ g$ and $h \circ h$ are p -morphisms of the form described in 6.8.6.

7.5.3. — One sees at once from the preceding description that if $\mathcal{R}$ and $\mathcal{R}'$ are two reduced root data of semisimple rank ≤ 2 , and one has a p -morphism from $\mathcal{R}'$ to $\mathcal{R}$, then $\mathcal{R}$ and $\mathcal{R}'$ are of the same type. More precisely, one has the following table.

Notations: Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a p -morphism. Denote by q (resp. q_1) any positive power of p . For systems of rank 2, use the notations of section 4 (denote the simple roots by α, β , with $\ell(\alpha) \leq \ell(\beta)$).

Type	p	values of f	values of ${}^t f$
Trivial	arbitrary	—	—
A_1	arbitrary	$f(\alpha') = q\alpha$	${}^t f(\alpha^*) = q\alpha'^*$
$A_1 \times A_1$	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q_1\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q_1\beta'^*$
A_2, B_2, G_2	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q\beta'^*$
B_2	$p = 2$	$f(\alpha') = q\beta$ $f(\beta') = 2q\alpha$	${}^t f(\alpha^*) = 2q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$
G_2	$p = 3$	$f(\alpha') = q\beta$ $f(\beta') = 3q\alpha$	${}^t f(\alpha^*) = 3q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$

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